HOCHSCHULE
RAVENSBURG-WEINGARTEN
UNIVERSITY
OF APPLIED SCIENCES

LIDAR AND RADAR SYSTEMS
GRADED HANDS-ON PROJECTS

Prof. Dr. Stefan Elser

GENERAL RULES AND INFORMATION

Compulsory Projects

Solve the task und understand your solution

In these slides, you will get a collection of all information needed to solve the tasks for the Hands-On projects.

During a written test, you will need to answers questions connected to these tasks. If you do not solve these tasks, you will not be able to answer these questions. This test will make for 50% of the overall grade.

The date of this additional test will be declared during the lecture.

OUR RWU DATASET
- LIDAR/RADAR/CAMERA

The RWU Dataset
The dataset

To get some experience with different sensor setups, we will work with our own dataset recorded at the RWU by Mr. Amoussou, Mr. Thaqi, Mr. Koschinski, Mr. Muzychuk and Mr. Berens.

In this dataset, we have access to data recorded by a full sensor set: camera, two different lidar types as well as a radar system.

The RWU Dataset
The sensors used

Let us take a closer look at the lidar sensors used:

- **Veldodyne Puck**
- **Blickfeld Cube 1**

As well as the radar sensor used:

- **MMWAVCAS-RF-EVM (Texas Instuments)**

The RWU Dataset
The dataset

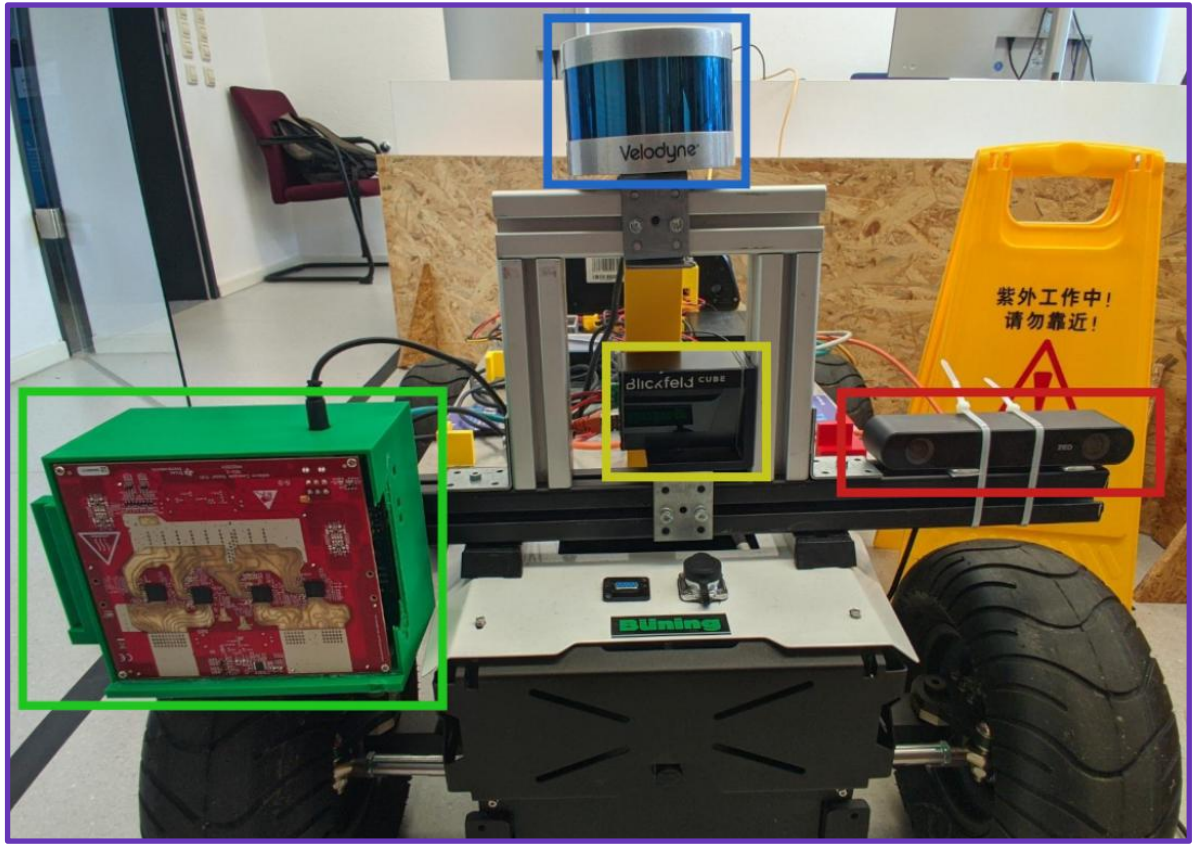
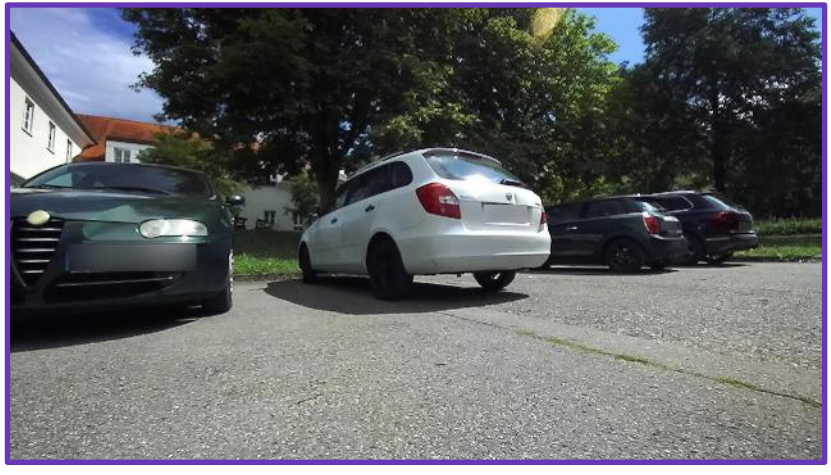


Fig.: All sensors used mounted on our robot.

Blue: Velodyne Puck
Yellow: Blickfeld Cube 1
Green: MMWAVCAS-RF-EVM
Red: ZED stereo camera

The RWU Dataset
3D sensor data



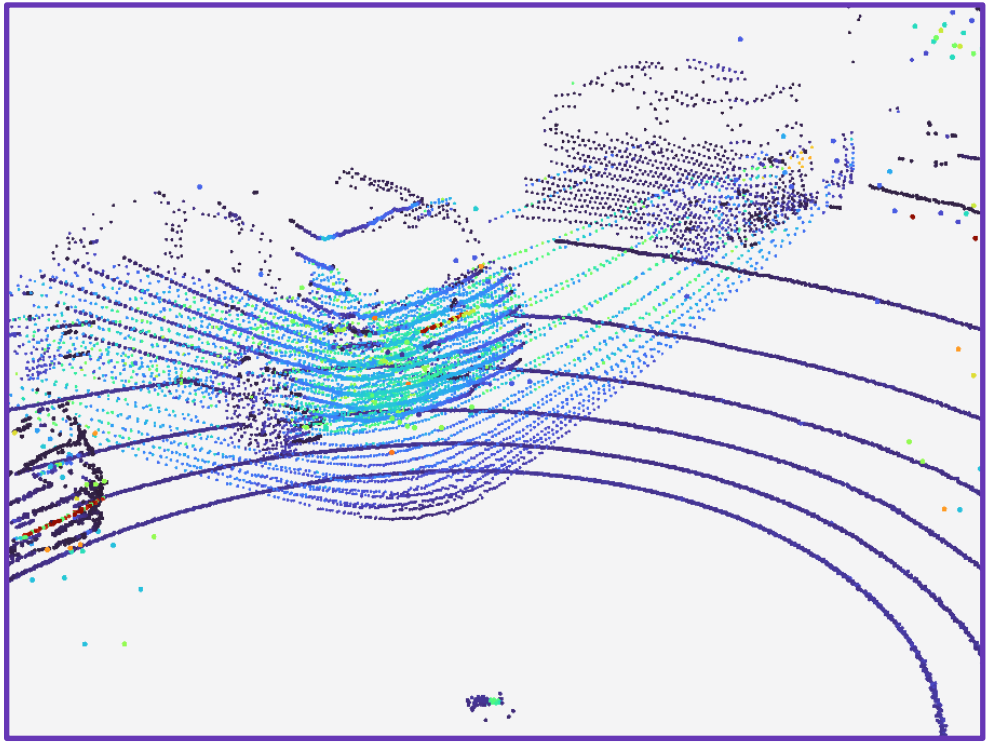
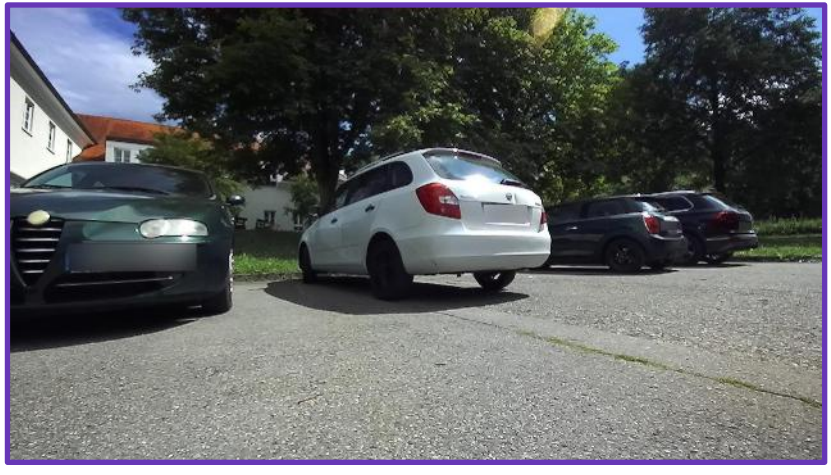


Fig.: A scene recorded at the same moment. Left: Camera image. Right: 3D view of all other sensor data combined into one coordinate system.

The RWU Dataset
3D sensor data



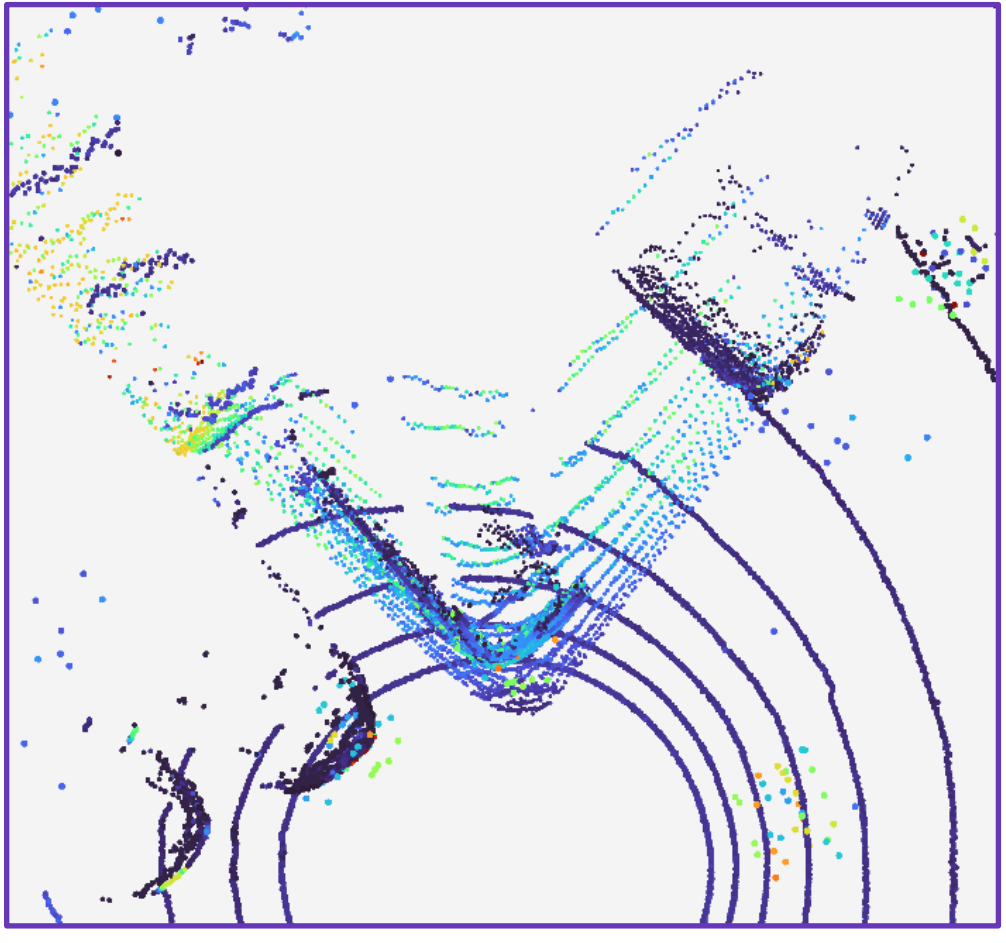


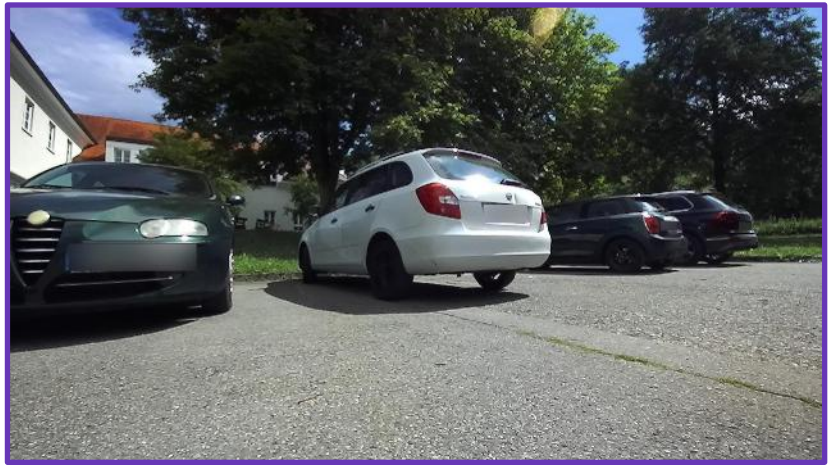
Fig.: A scene recorded at the same moment. Left: Camera image. Right: Bird's eye view (BEV) of all other sensor data combined into one coordinate system.

The RWU Dataset
Velodyne Puck



Fig.: Velodyne Puck, a rotating 360°lidar system.
Source: https://velodynelidar.com/products/puck/

The RWU Dataset
Velodyne Puck



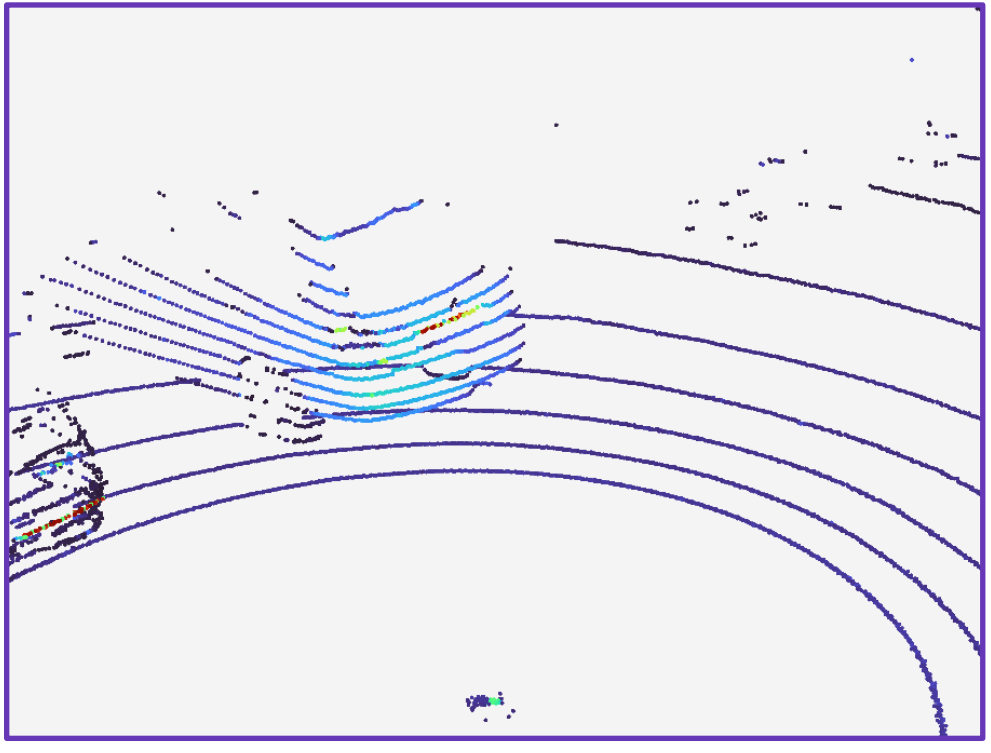
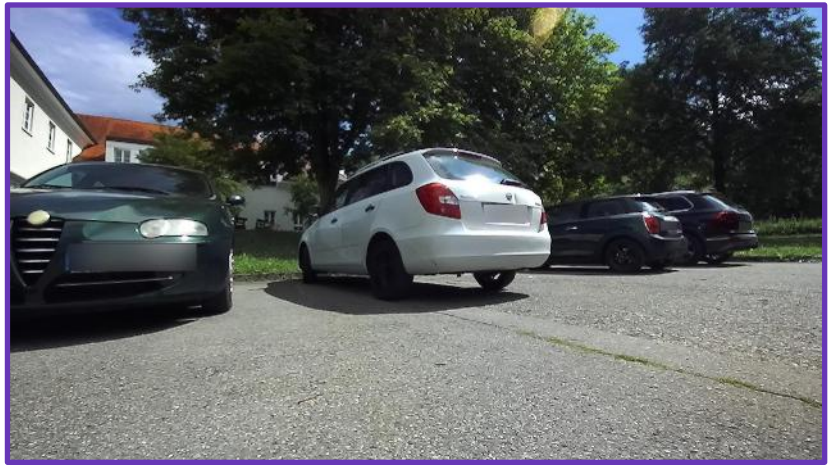


Fig.: A scene recorded at the same moment. Left: Camera image. Right: Velodyne Puck point cloud.

The RWU Dataset
Velodyne Puck



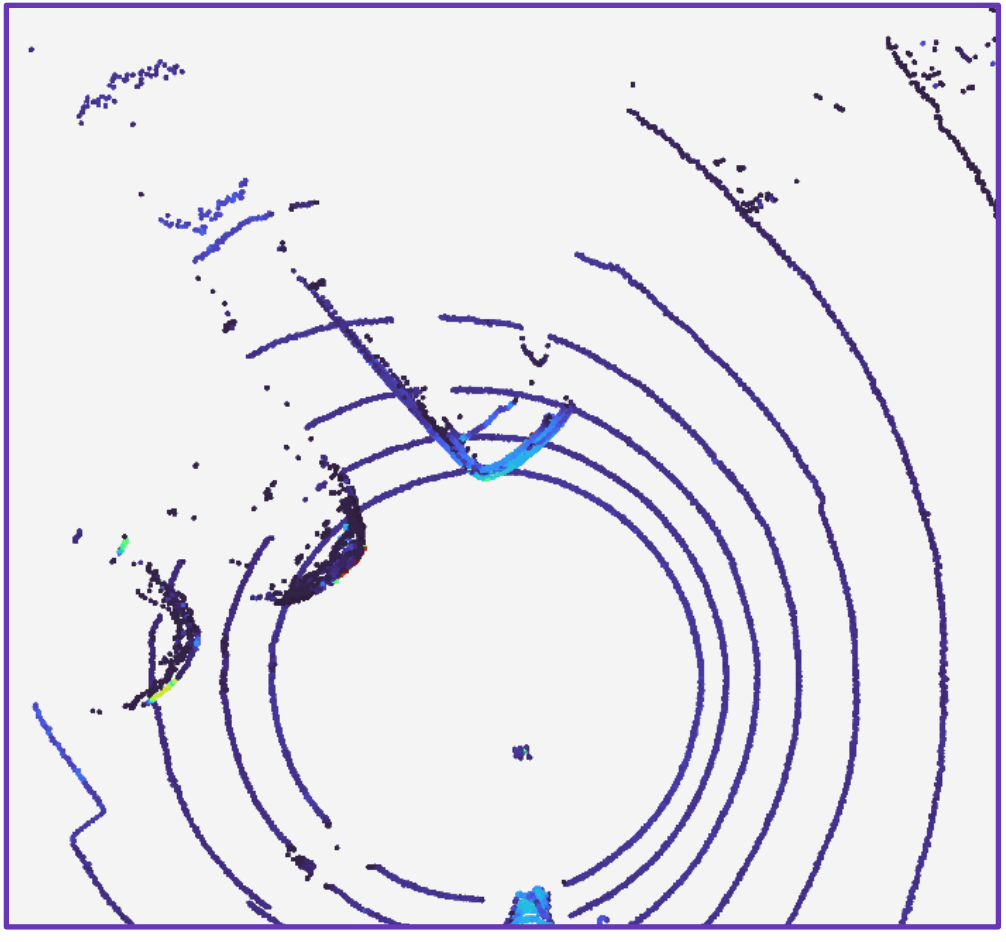


Fig.: A scene recorded at the same moment. Left: Camera image. Right: BEV of Velodyne Puck point cloud.

The RWU Dataset
Blickfeld Cube 1

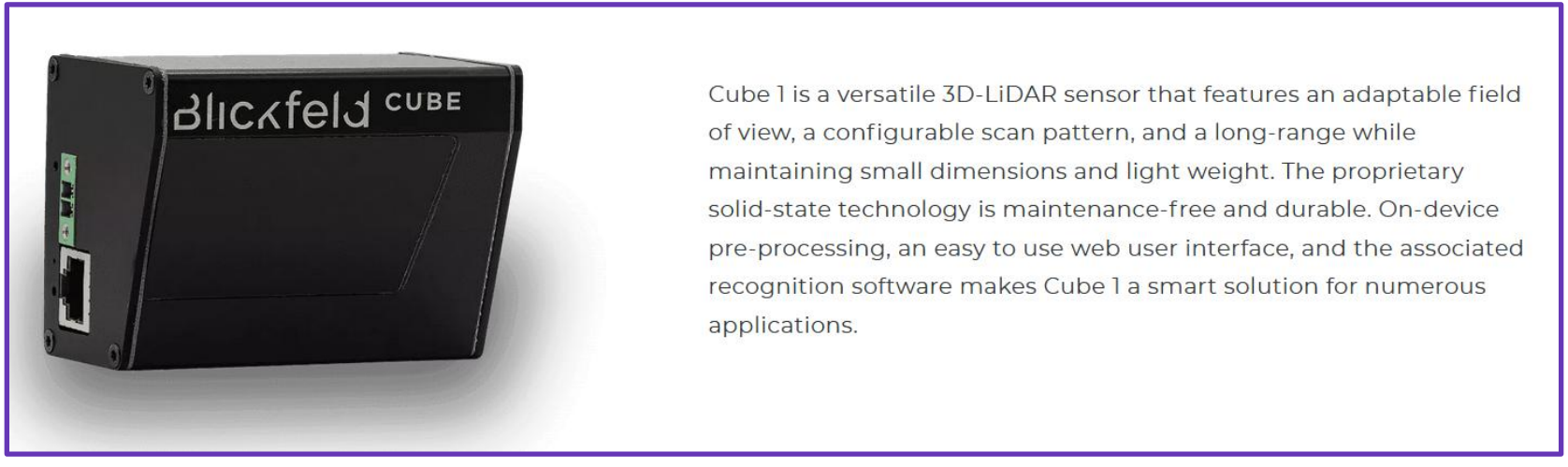


Fig.: Blickfeld Cube 1, a solid state lidar system.
Source: https://www.blickfeld.com/products/cube-1/

The RWU Dataset
Blickfeld Cube 1



Fig.: Blickfeld Cube 1, a solid state lidar system. Source: https://www.blickfeld.com/products/cube-1/

Key Specifications

Typical application range	1.5 – 75 m
Maximum detection range	250 m
Field-of-view (H x V)	70° x 30°
Vertical resolution	5 – 400 scan lines per frame (user-configurable)
Horizontal resolution	0.4° – 1.0° (user-configurable)
Operating voltage	12 V
Power consumption	less than 15 W
Laser class	Class 1, eye-safe (IEC 60825-1:2014, Ed. 3)

The RWU Dataset
Blickfeld Cube 1

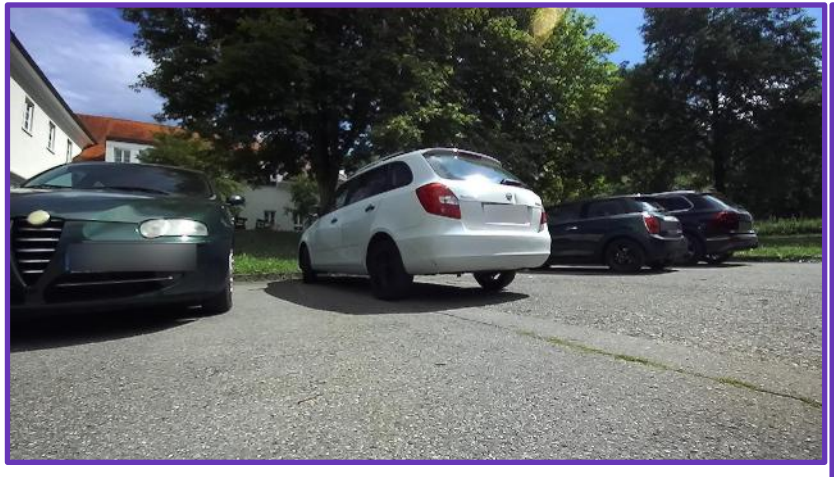


Fig.: Blickfeld Cube 1, a solid state lidar system. Source: https://www.blickfeld.com/products/cube-1/

Key Specifications

Typical application range	1.5 – 75 m
Maximum detection range	250 m
Field-of-view (H x V)	70° x 30°
Vertical resolution	5 – 400 scan lines per frame (user-configurable)
Horizontal resolution	0.4° – 1.0° (user-configurable)
Operating voltage	12 V
Power consumption	less than 15 W
Laser class	Class 1, eye-safe (IEC 60825-1:2014, Ed. 3)

The RWU Dataset
Blickfeld Cube 1



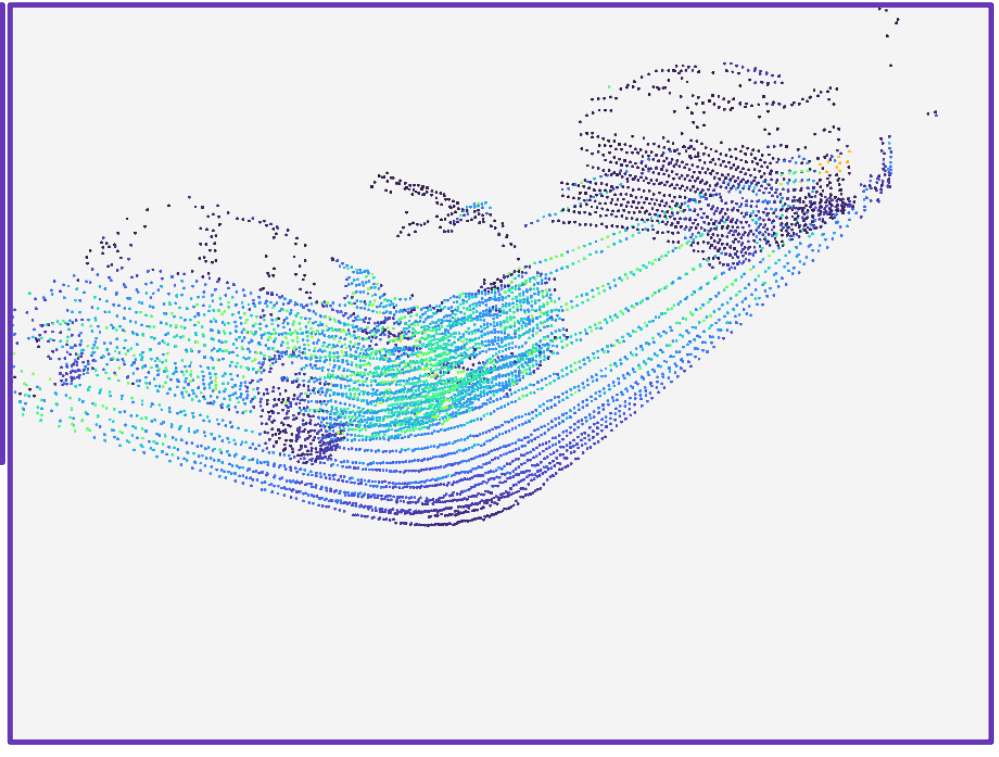
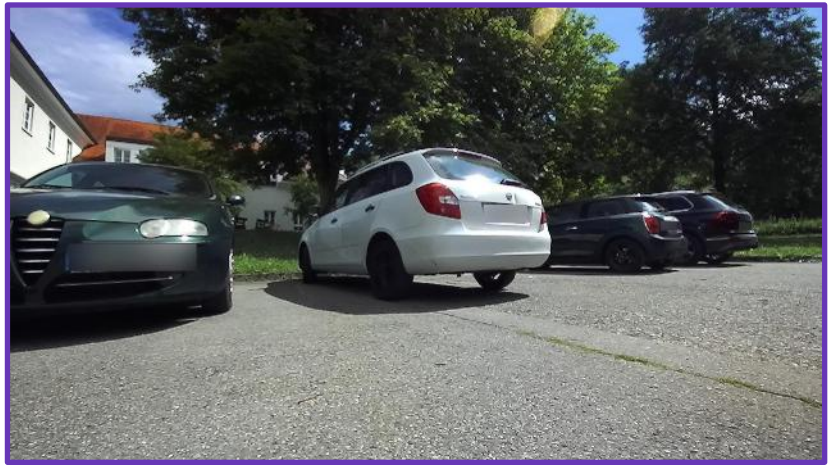


Fig.: A scene recorded at the same moment. Left: Camera image. Right: Blickfeld Cube 1.

The RWU Dataset
Blickfeld Cube 1



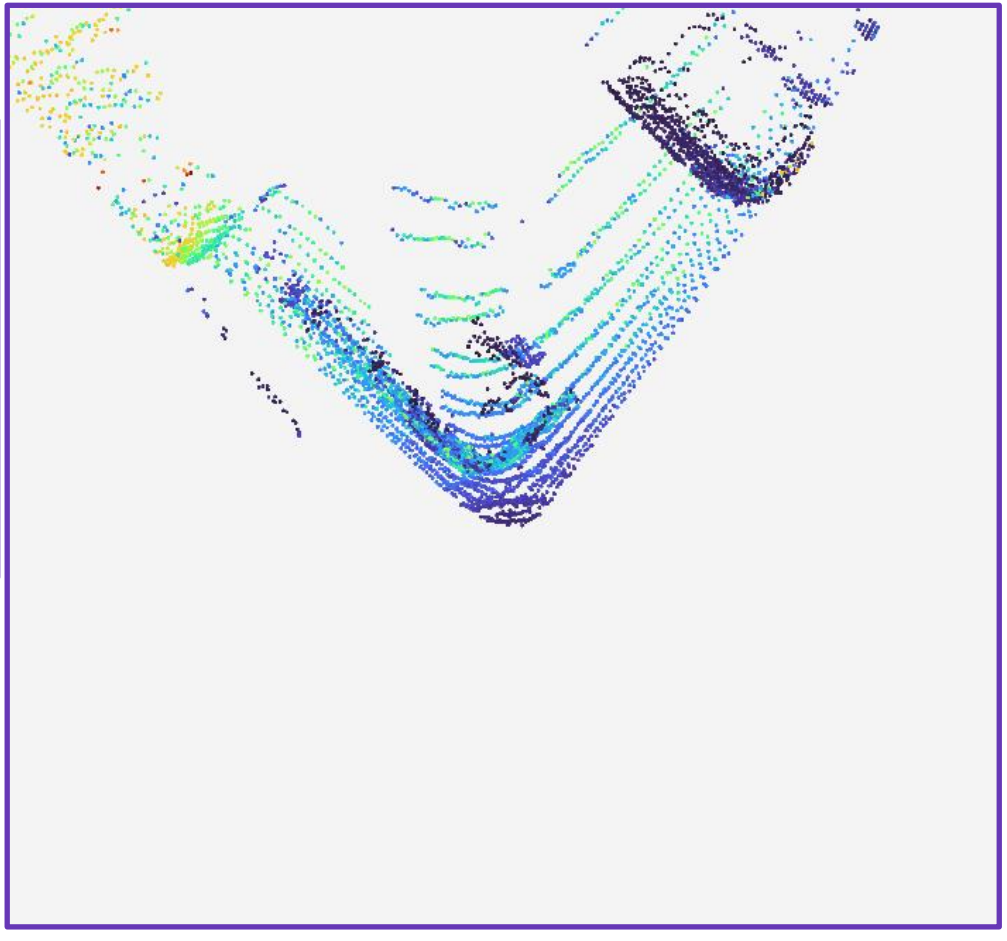


Fig.: A scene recorded at the same moment. Left: Camera image. Right: BEV of Blickfeld Cube 1 point cloud.

The RWU Dataset
MMWCAS-RF-EVM

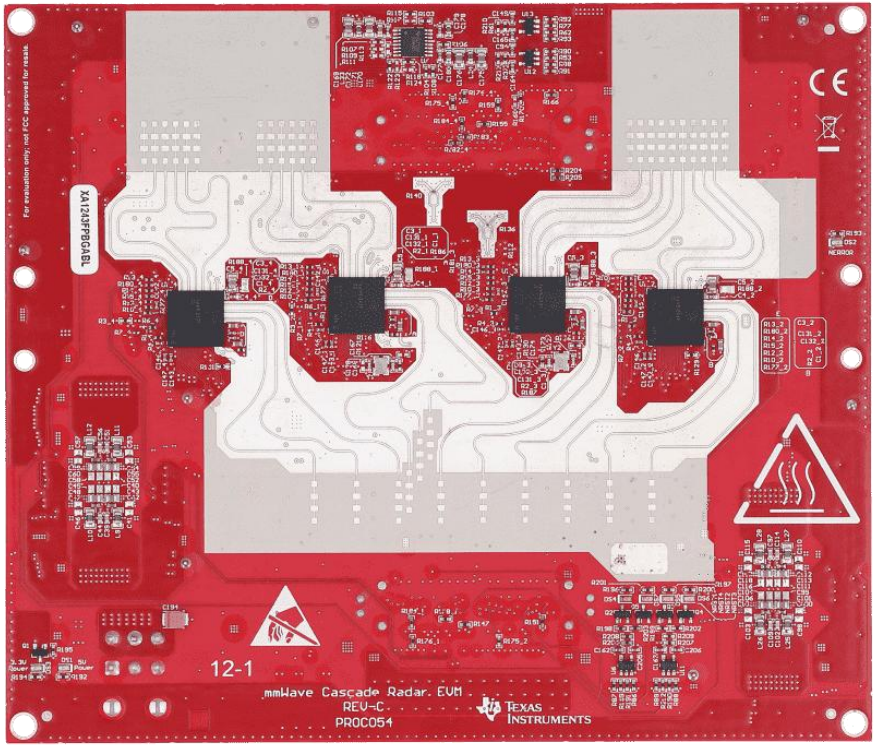


Fig.: The AWRx Cascaded Radar RF Evaluation Module (MMWCAS-RF-EVM) from Texas Instruments.
Source: https://www.ti.com/tool/MMWCAS-RF-EVM

The RWU Dataset
MMWCAS-RF-EVM

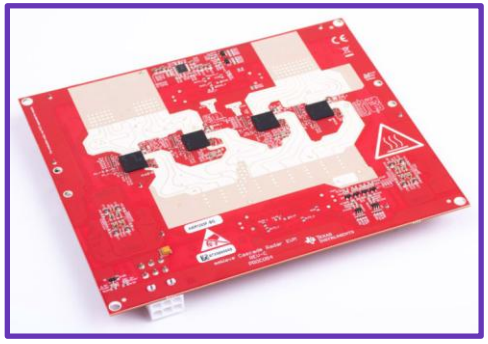


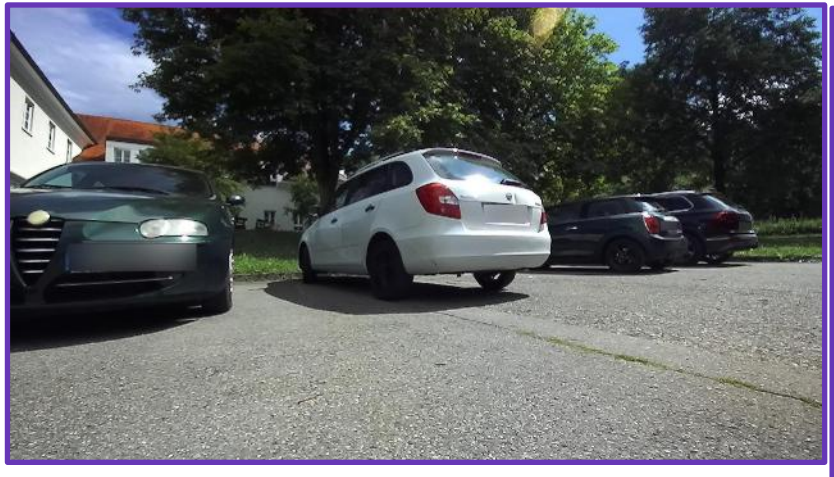
Fig.: Key features of the MMWCAS-RF-EVM as listed in the documentation by Texas Instruments.

Copyright and source: Texas Instruments

Table 1. Key Features

	Cascade Radar RF Board
4x AWRx 76-81GHz Radar SoC	Integrated VCO, LO distribution, PA, LNA, ADC, 3 TX and 4 RX ARM MCU R4 Controller
AWRx RF/LO Peripherals	
12x TX, 16x RX Antennas	• 12 total transmitters across all 4 AWRx devices • 16 total receivers across all 4 AWRx devices • 86 non-overlapping Azimuth virtual array (lambda/2) • 4 minimum redundancy array (MRA) Elevation
Embedded Antenna	• 4-element series-fed patch antenna, 12dBi • ± 60 deg Azimuth 3dB • ± 30 deg Elevation 3dB
20 GHz LO Star-Network Distribution	2x Wilkinson power dividers fed by the Master AWRx device LO output to Master and Slave AWRx devices
AWR Digital Peripherals	
CSI2.0 4-lane	600Mbps/Lane for 2.4 Gbps ADC IF data per device
QSPI Flash	16 Mbit QSPI flash for AWR firmware updates/deveopment
Serial Peripherals	SPI, I2C, UART, GPIO
System Temperature	TMP112 I2C Temperature Sensors
Power	
Radar Power Management IC (PMIC) Solution	• 2x LP87524P Quad-Channel, Integrated FET, Buck Converters, secondary LC filtering solution powering all 4 AWRx devices • TPS73733 LDO generating 3.3V system power

The RWU Dataset
MMWCAS-RF-EVM



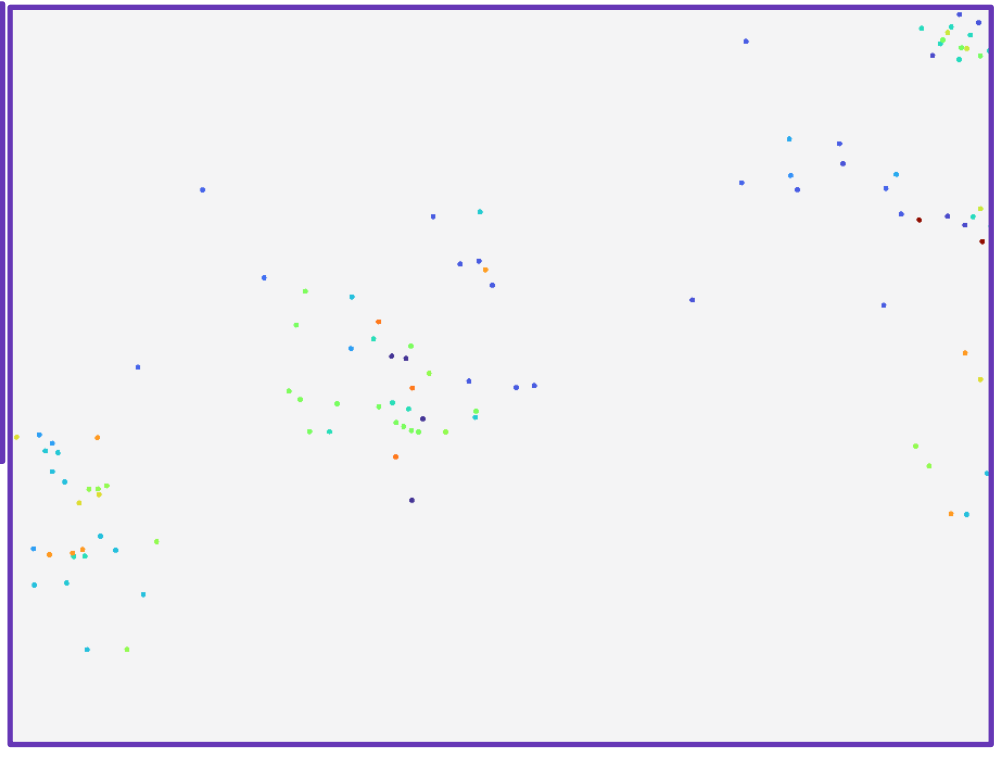
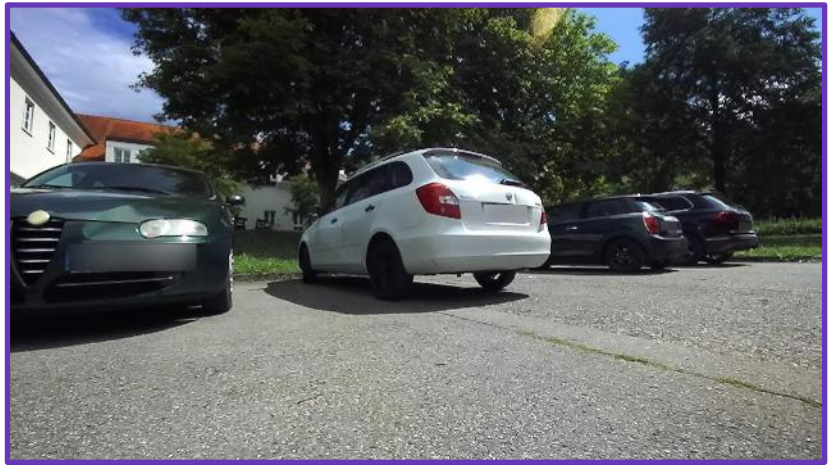


Fig.: A scene recorded at the same moment. Left: Camera image. Right: MMWCAS-RF-EVM.

The RWU Dataset
MMWCAS-RF-EVM



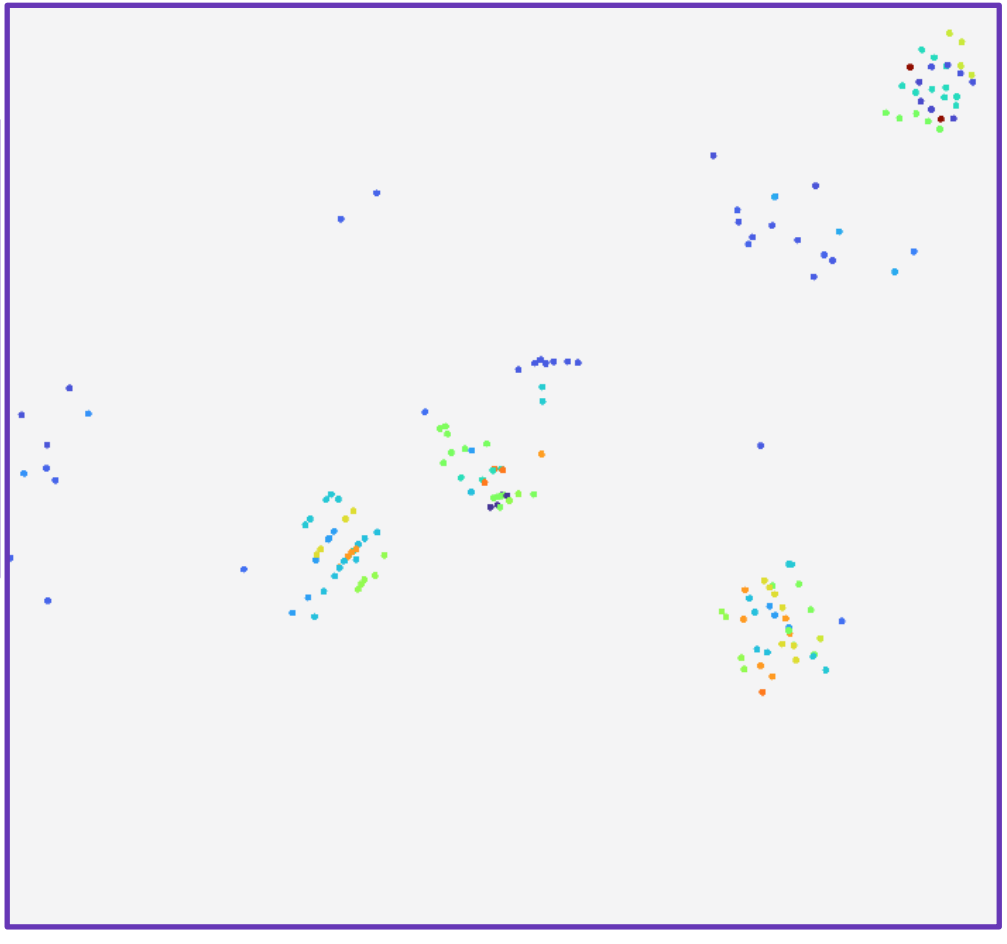


Fig.: A scene recorded at the same moment. Left: Camera image. Right: BEV of MMWCAS-RF-EVM point cloud.

The RWU Dataset
Getting access to the dataset

It will be best to take a look at the dataset yourselves before you head into the tasks defined below. For this, follow the following steps:

1. **Download und unzip the dataset:**
   The 28GB dataset is available here:
   https://bwsyncandshare.kit.edu/s/spbj9pNqfRm43eG

2. **(optional) Use foxglove to visualize the data:**
   You can visualize the data without any coding using the software foxglove:
   https://foxglove.dev/

The RWU Dataset
Using the data

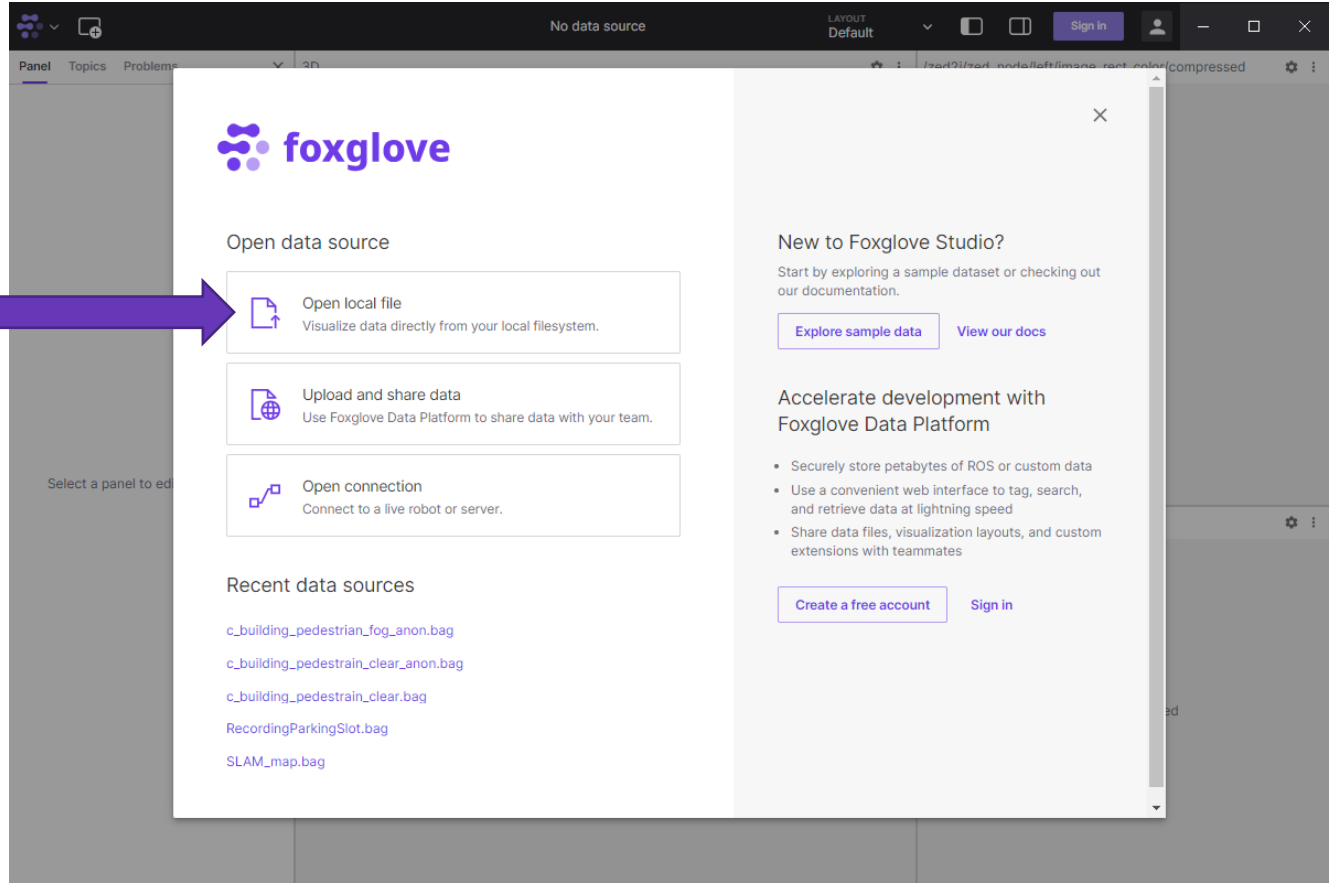


Fig.: Starting screen of foxglove.

Here, you can open the bag files of our dataset.

The RWU Dataset
Using the data

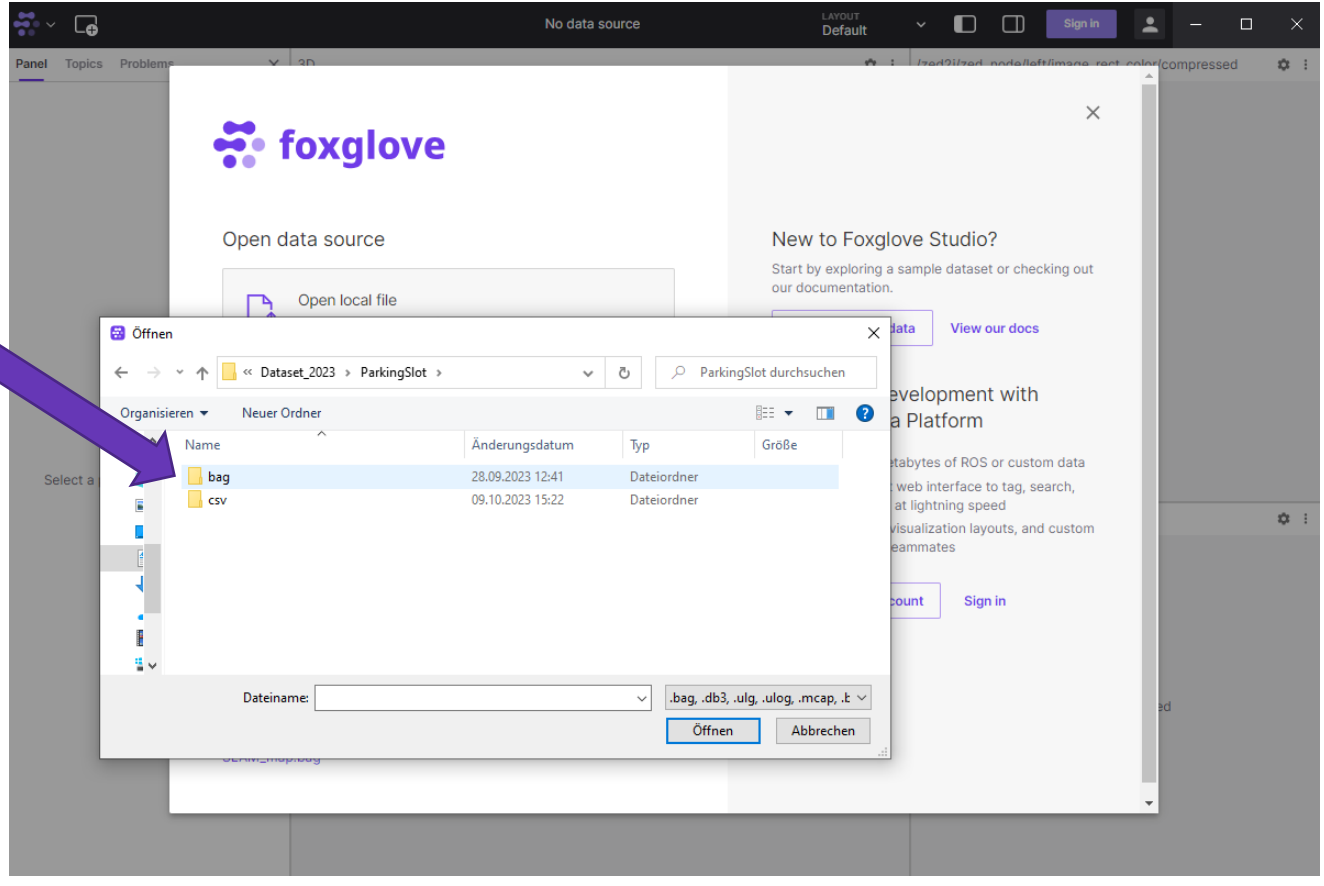


Fig.: Choose the scenario and look for the bag subfolder, e.g.

Dataset_2023\ParkingSlot\bag

The RWU Dataset
Using the data

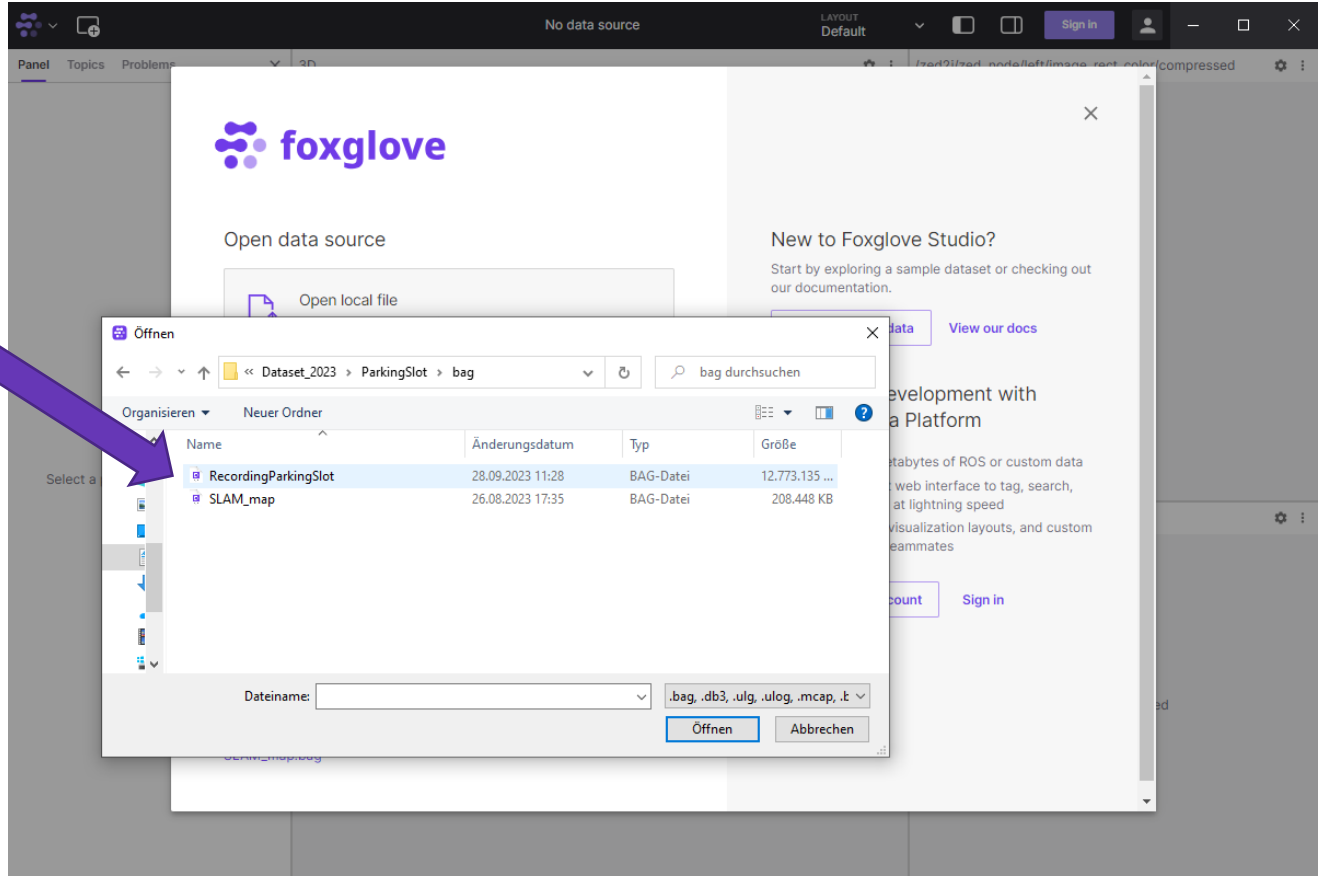


Fig.: Choose a bag file and take a look.

The RWU Dataset
Using the data

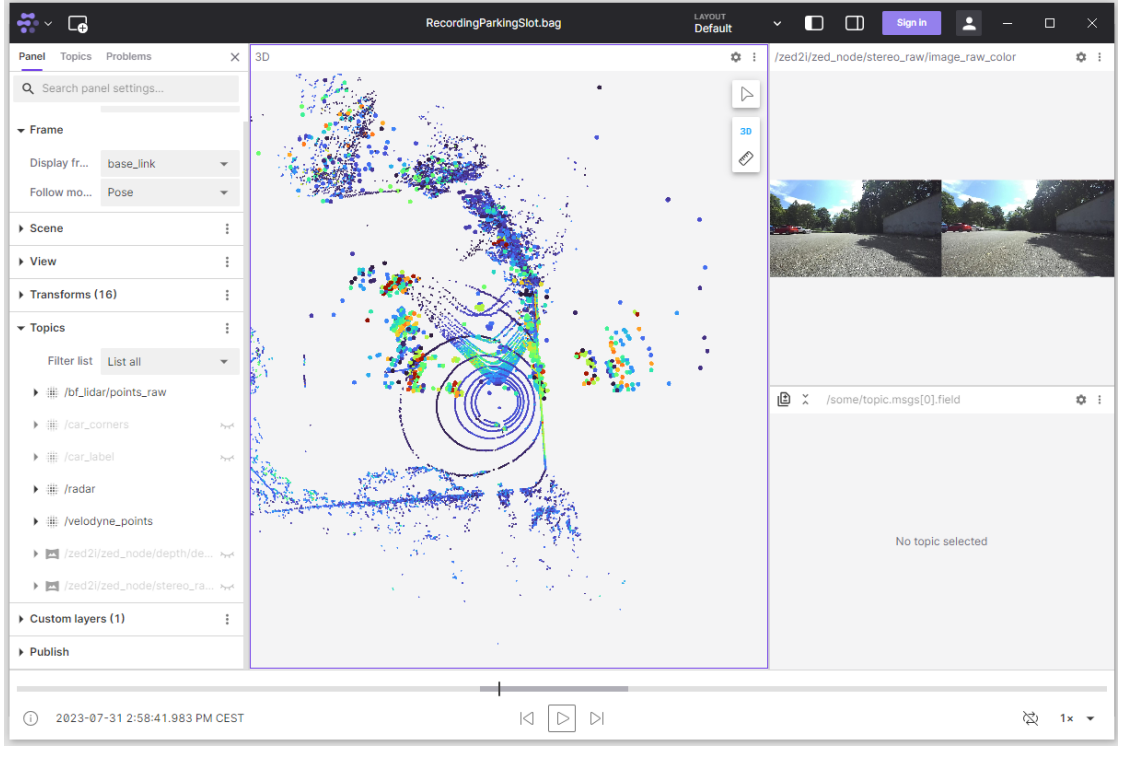


Fig.: Playback of RecordingParkingSlot.bag

In each sub-window, you can choose what is to be visualized. Depending on the data you want to open, some settings need to be changed as well.

Just give it a try to see how this works.

The RWU Dataset
Using the data

1. **Download und unzip the dataset**
2. **(optional) Use foxglove to visualize the data**
3. **Read the pdf “RWU Dataset” provided via Moodle**

ACHTUNG:

You will have to work with the .csv files and not the .bag files. So foxglove is really only used for the visualization before you begin to code.

TASK 1:
INFLUENCE OF FOG ON THE SENSORS

Influence of Fog on the Sensors
Data to be used

We want to evaluate the influence of fog on the sensors.

For this, the team has recorded data inside of C004 which was then flooded using a fog machine.



Influence of Fog on the Sensors

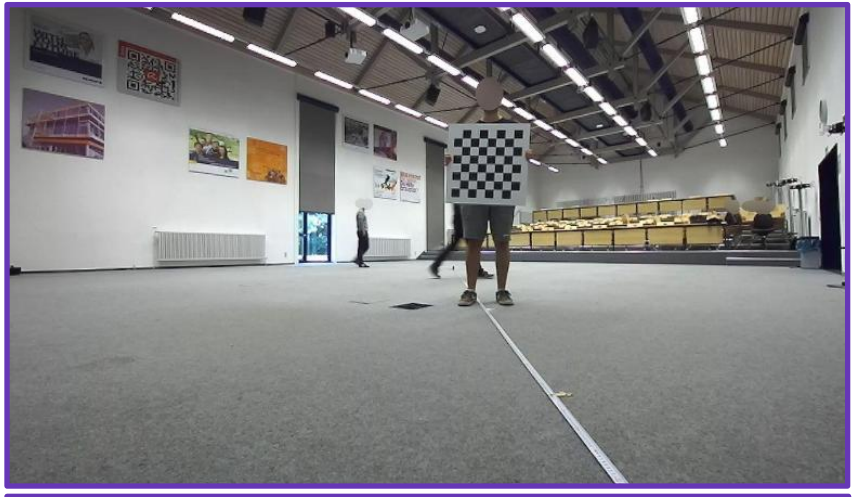
Data to be used

Obviously, fog has a severe influence on the camera images as seen on the right.

But what about the lidar or radar sensors?

Top:
c_building_pedestrain_clear_anon

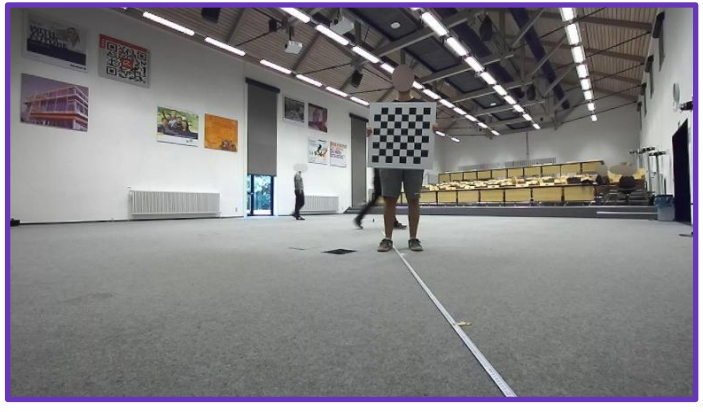
Bottom:
c_building_pedestrian_fog_anon





Influence of Fog on the Sensors

Data to be used



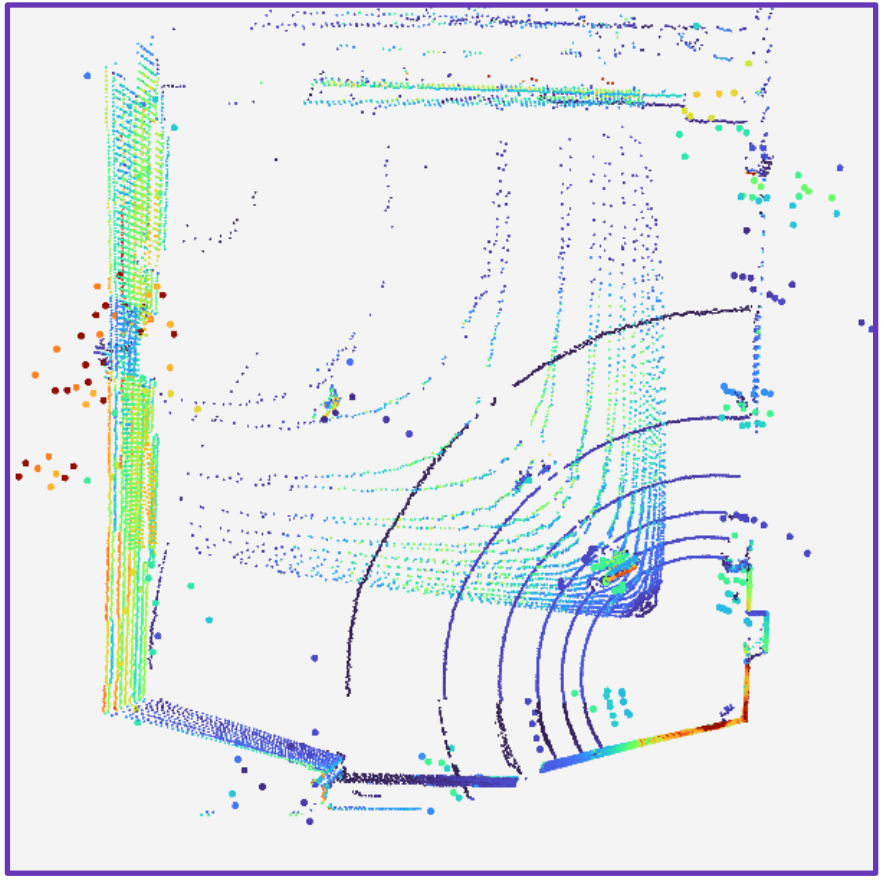


Fig.: BEV of all sensor data in the clear scene “c_building_pedestrian_clear_anon” provided in the dataset.

Influence of Fog on the Sensors

Data to be used



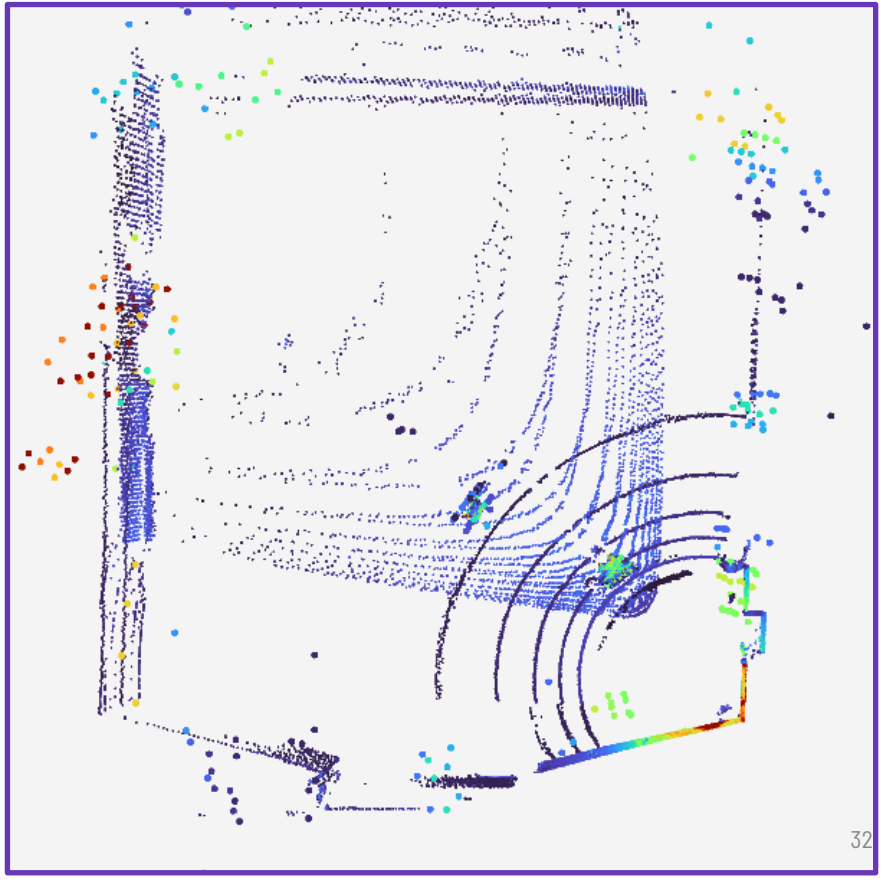


Fig.: BEV of all sensor data in the foggy scene “c_building_pedestrian_fog_anon” provided in the dataset.

Influence of Fog on the Sensors

Data to be used

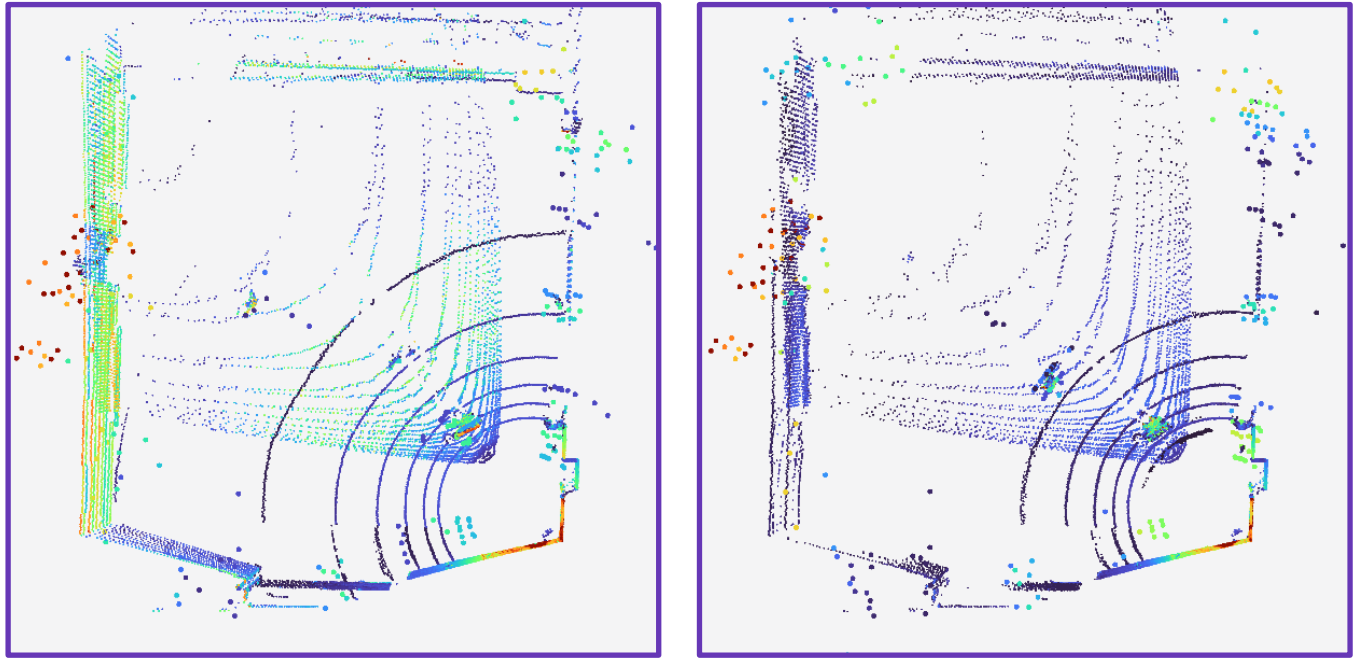


Fig.: Left: Clear Scene. Right: Scene filled with fog.

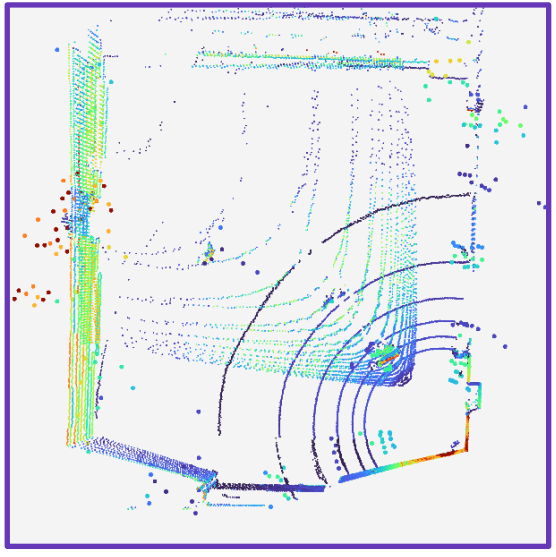
Take a close look at the data. Here, the color represents the intensity, how strong the reflections are. Note that the intensity drops drastically.

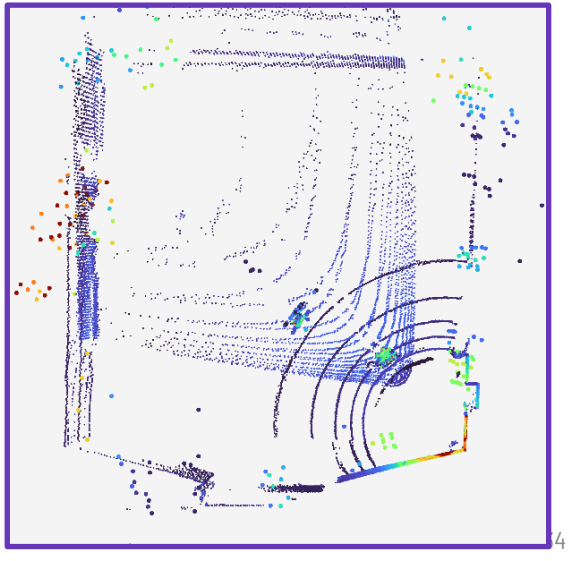
Influence of Fog on the Sensors
Data to be used

We now want to evaluate how the sensors are influenced by fog.

Obviously, the sensors are influenced differently. So we need to provide an evaluation for each sensor separately.

Taking a look at the Velodyne data, we realize that a lot of points seem to be missing...





Influence of Fog on the Sensors

Data to be used

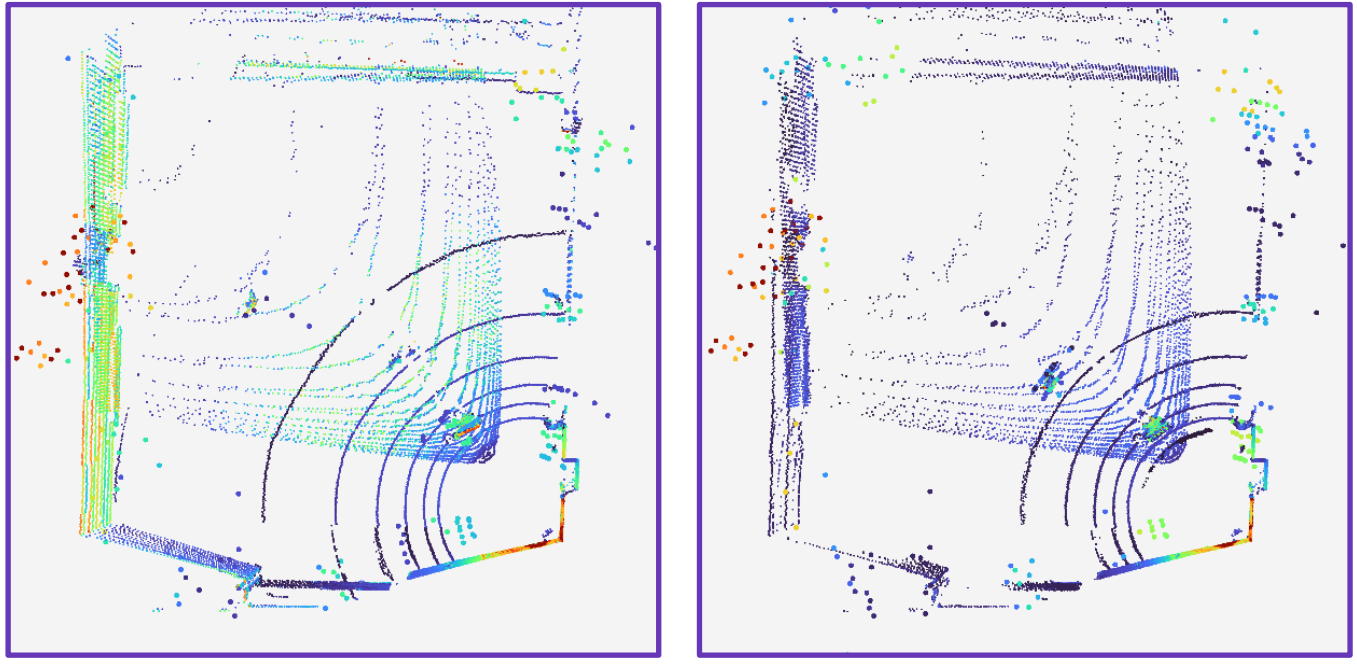


Fig.: Left: Clear Scene. Right: Scene filled with fog. While the Blickfeld data only seems to drop in intensity, the Velodyne is already missing a full scan line....

Influence of Fog on the Sensors

Data to be used

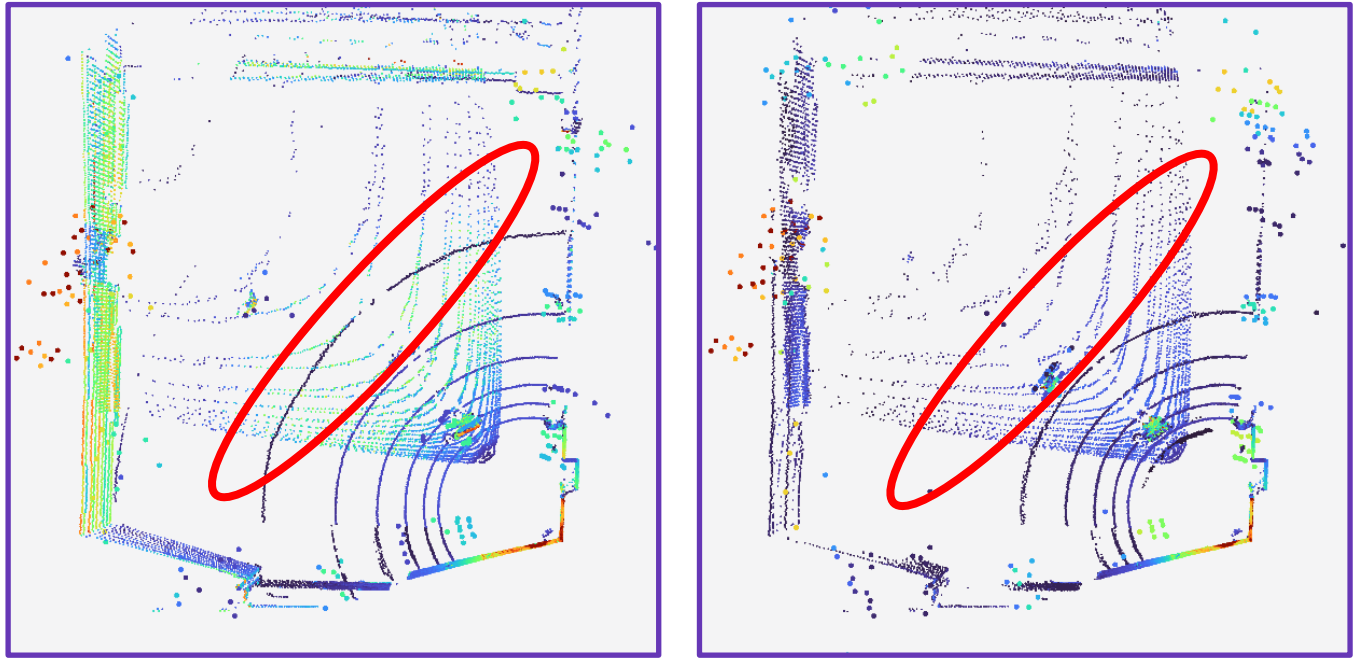


Fig.: Left: Clear Scene. Right: Scene filled with fog. While the Blickfeld data only seems to drop in intensity, the Velodyne is already missing a full scan line marked in red.

Influence of Fog on the Sensors

Data to be used

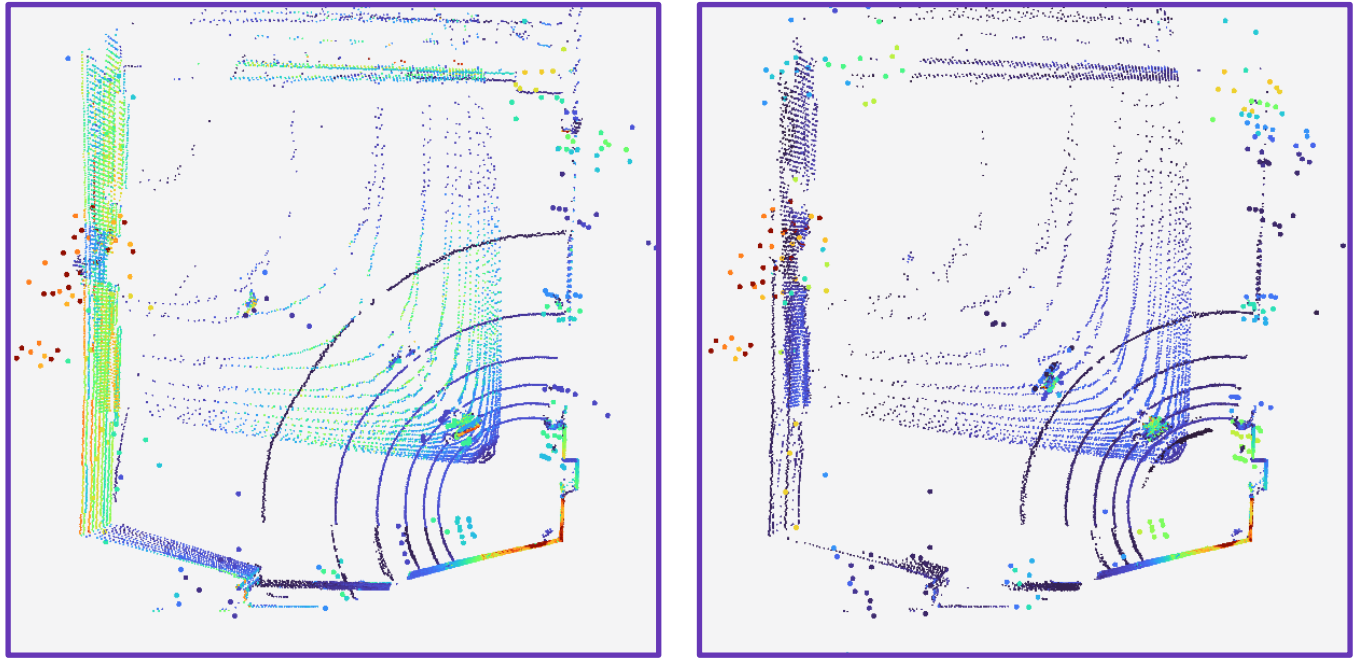


Fig.: On the other hand, the fog seems to generate points in close range ...

Influence of Fog on the Sensors

Data to be used

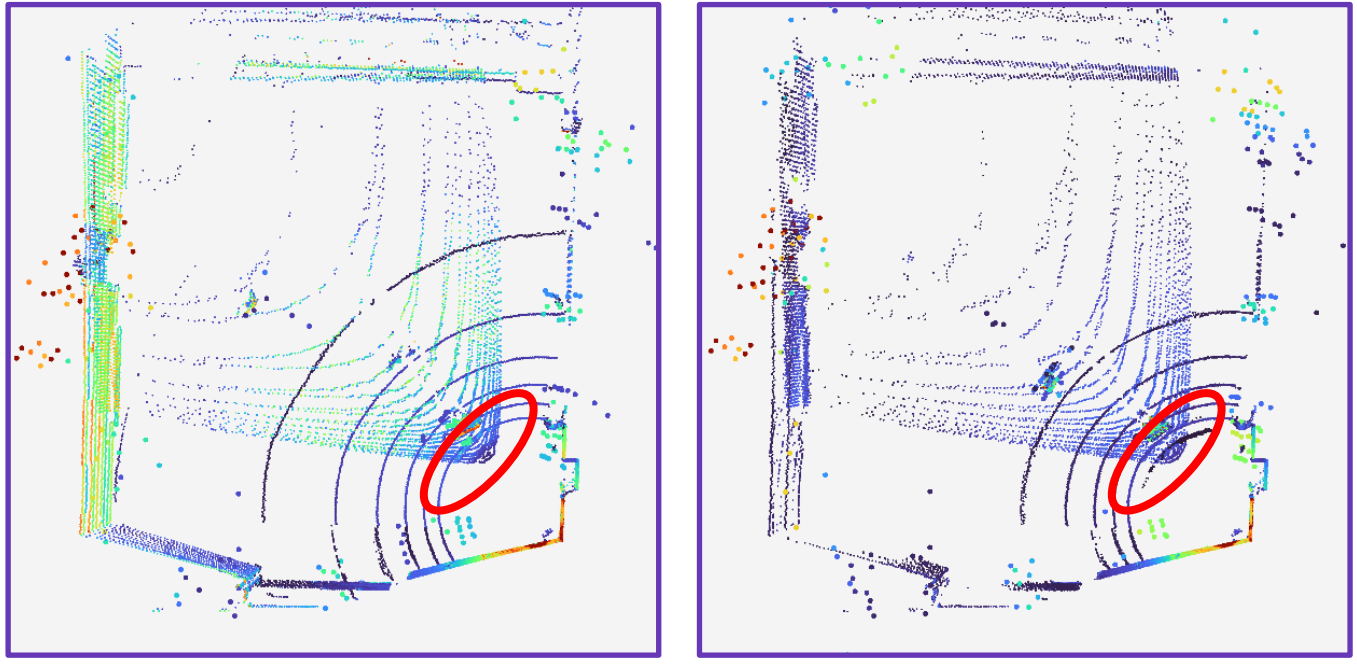


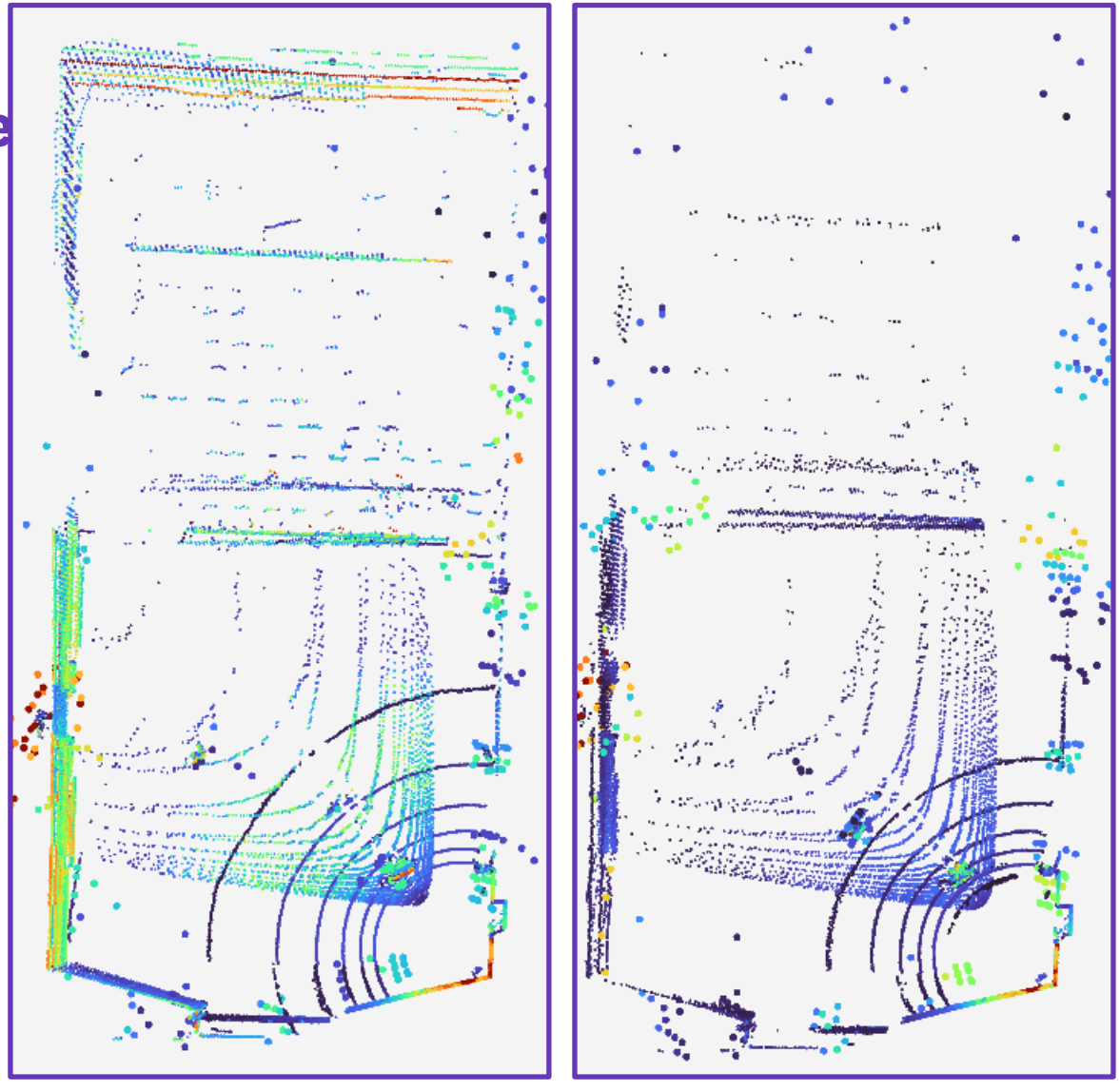
Fig.: On the other hand, the fog seems to generate points in close range again marked in red. These actually come from the Blickfeld sensor and not from the Velodyne.

Influence of Fog on the
Data to be used

Fig.: If we zoom out, we realize that both lidar sensors are not able to measure in far range as good as in a clear scene.

The radar sensor on the other hand seems to have no problem with the fog.

Left: Scene without fog.
Right: Scene filled with fog.



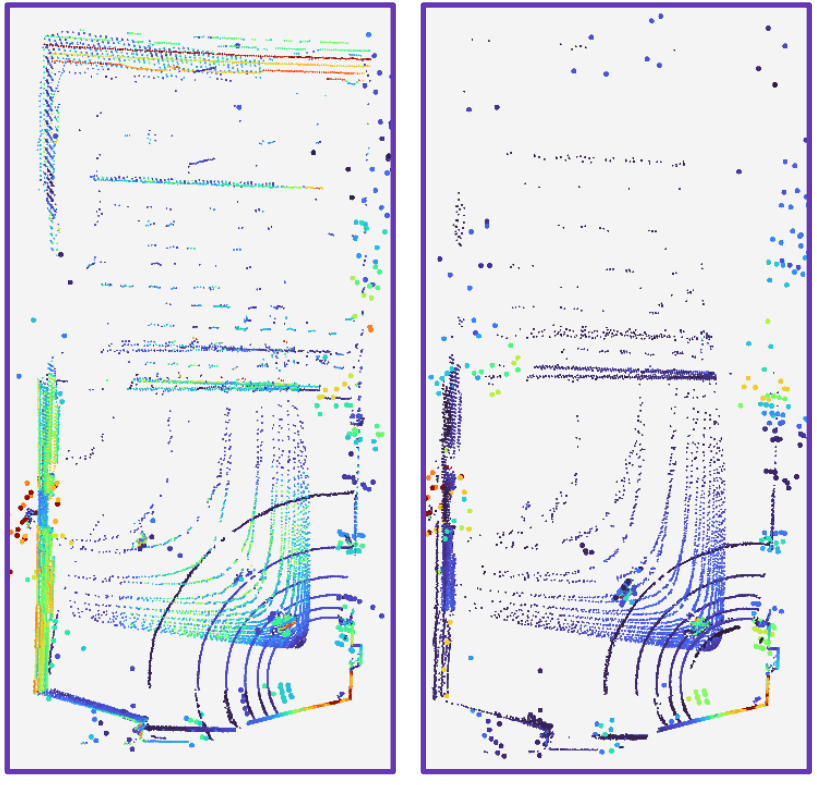
Influence of Fog on the Sensors
Data to be used

For each sensor, we want to evaluate the influence of fog.

For this, we will use the .csv files in the following folders:

c_building_pedestrian_clear_anon

c_building_pedestrian_fog_anon

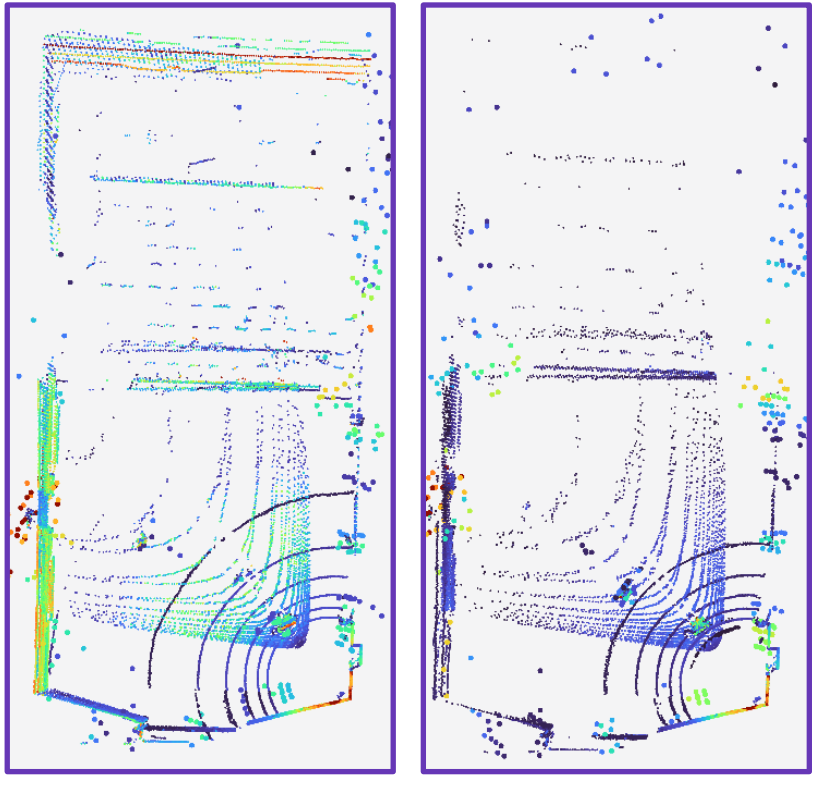


Influence of Fog on the Sensors
Data to be used

Just to make sure that we are all on the same page here:

Use the csv files for the tasks!

The bag files are only there, if you want to take a quick look at the data.

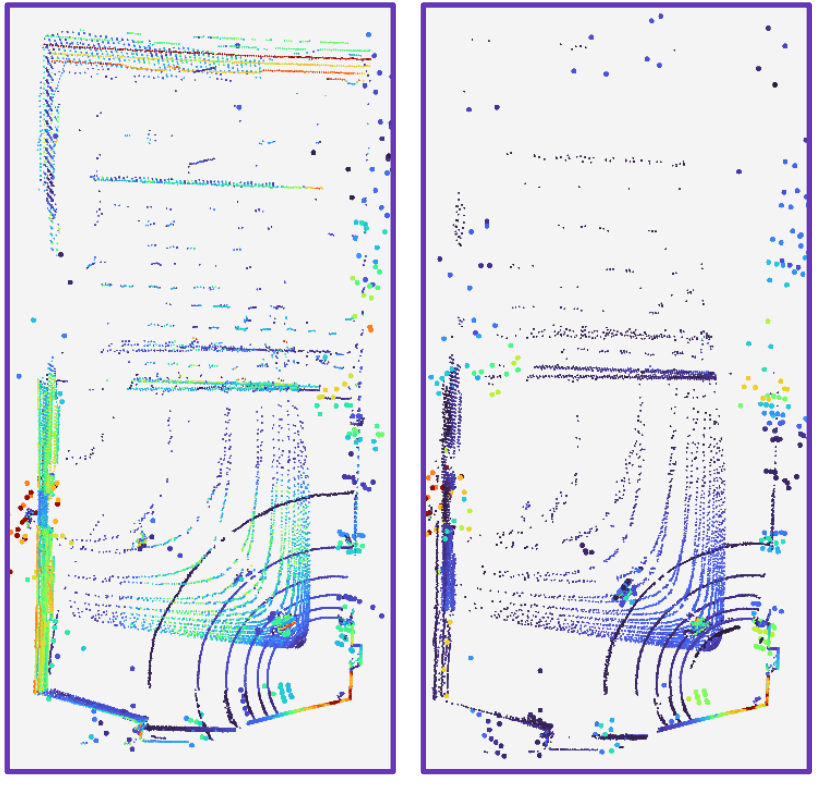


Influence of Fog on the Sensors

Data to be used

As seen above, the distance will have a heavy influence whether or not a point is detected - at least for the two lidar sensors.

Each point is provided with 3 coordinates $(x, y, z) \in \mathbb{R}^3$ in the same coordinate system, so we can easily calculate its distance.

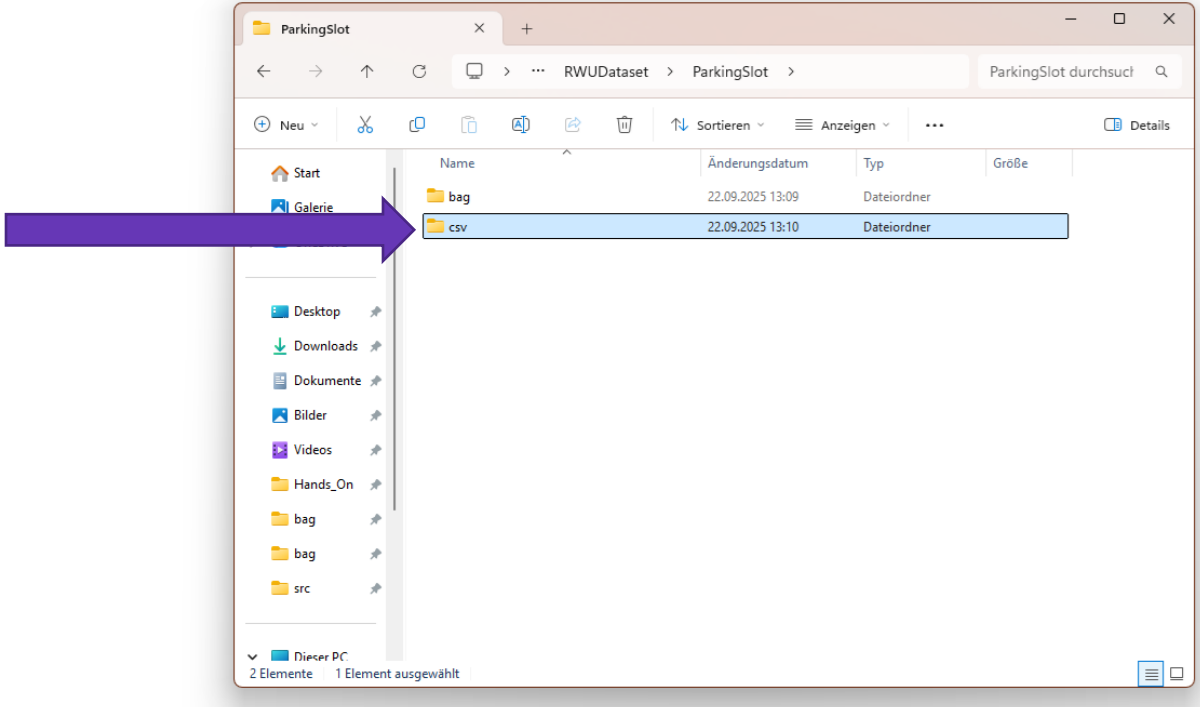


Influence of Fog on the Sensors

Data to be used

Fig.: Choose the csv files. There, you will find for each sensor the data stored as csv.

The first three entries correspond to the values $(x, y, z) \in \mathbb{R}^3$

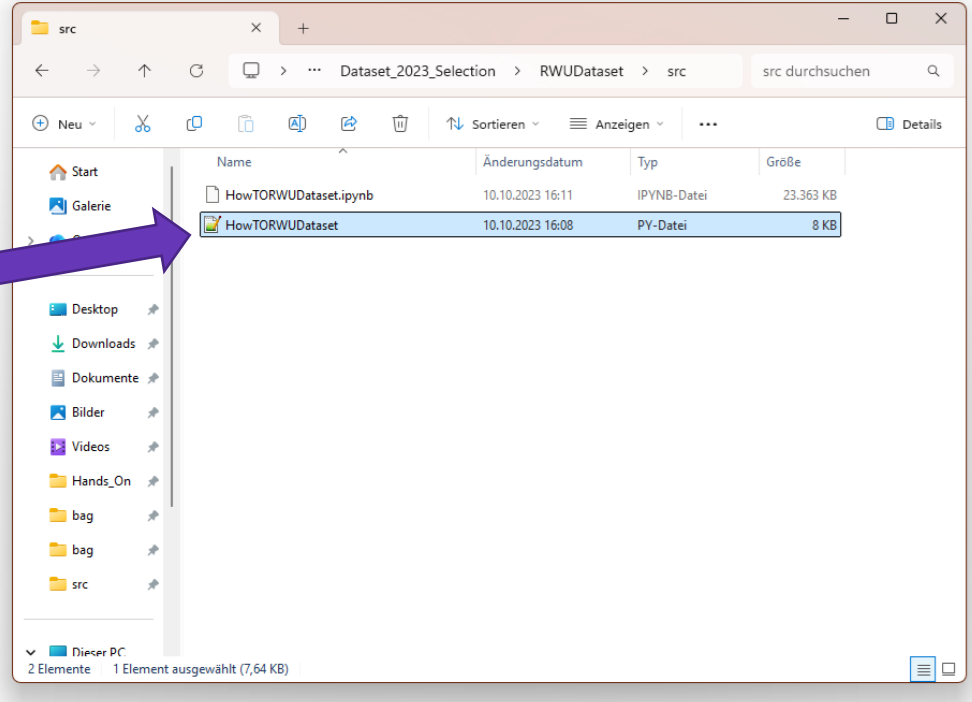


Influence of Fog on the Sensors

Data to be used

Fig.: In src, you will find source code which explains how to use the data.

This is provided as a python file as well as a jupyter notebook



Influence of Fog on the Sensors
Data to be used

As a possible way to evaluate each sensor, we now could sort all measurements during the recording using the distance and calculate a histogram (as seen on the right).

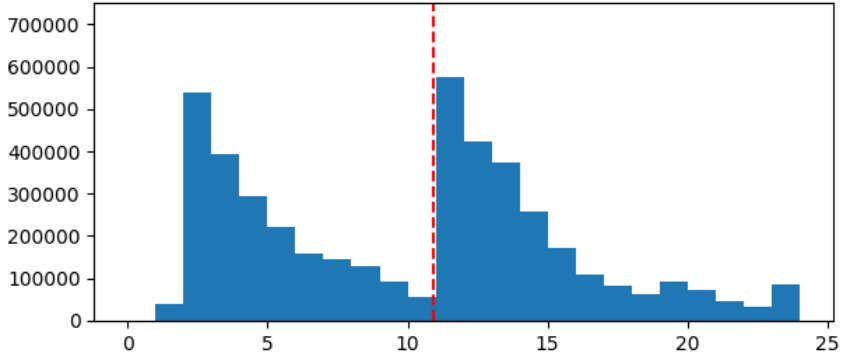


Fig.: Example of a histogram.

For each sensor, we do this for the two recordings:

c_building_pedestrian_clear_anon

c_building_pedestrian_fog_anon

Influence of Fog on the Sensors
Data to be used

What we have to do:

Evaluate the influence of fog on the sensors. For this task, we will compare the following recordings using the csv files:

c_building_pedestrian_clear_anon

c_building_pedestrian_fog_anon

You need to decide on how to evaluate the data! Make sure to understand the process you have decided on. Also understand why there is an influence if you measured any.

Task 1: Influence of Fog on the Sensors
Goals

Goals of this task:

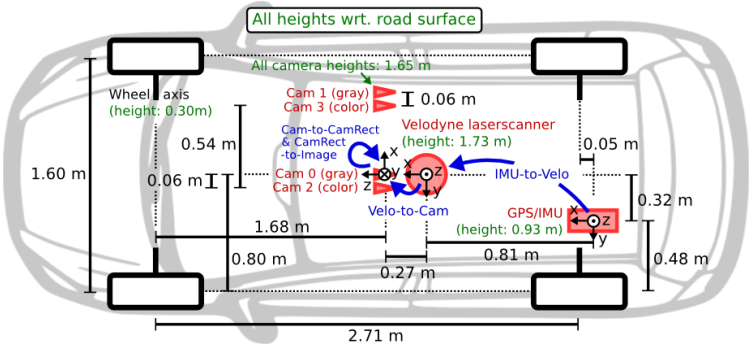
- Read the pdf “RWU Dataset” provided via Moodle and download the dataset.
- Evaluate the influence of fog on the sensors. Understand why there is an influence if you measured any.
- Make sure to fully understand the process you have used.

TASK 2:
SENSOR FUSION

Sensor Fusion
What we have to do

During this task, we want to implement an object detector in a 3D environment using camera and lidar data.

For this task, we will work with a subset of the KITTI dataset in a reduced format.





Sensor Fusion
Camera based object detection

We want to use a precise object detection in the camera system. For this, the YOLOv8 object detector is a reasonable choice.

This can be used quite easily by pip install:

https://pypi.org/project/ultralytics/

https://docs.ultralytics.com/

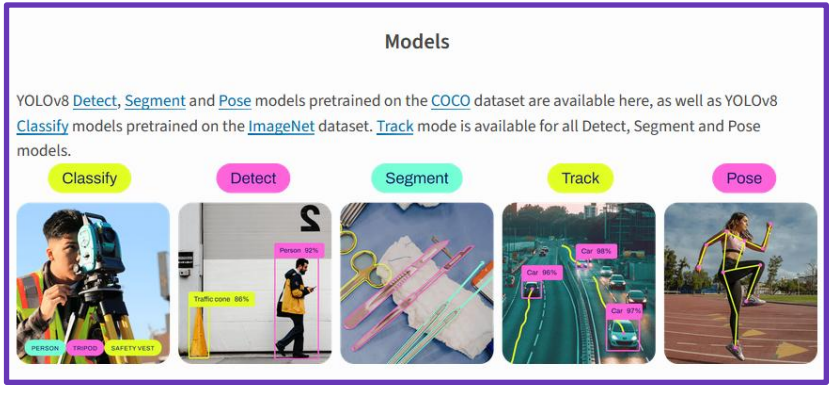
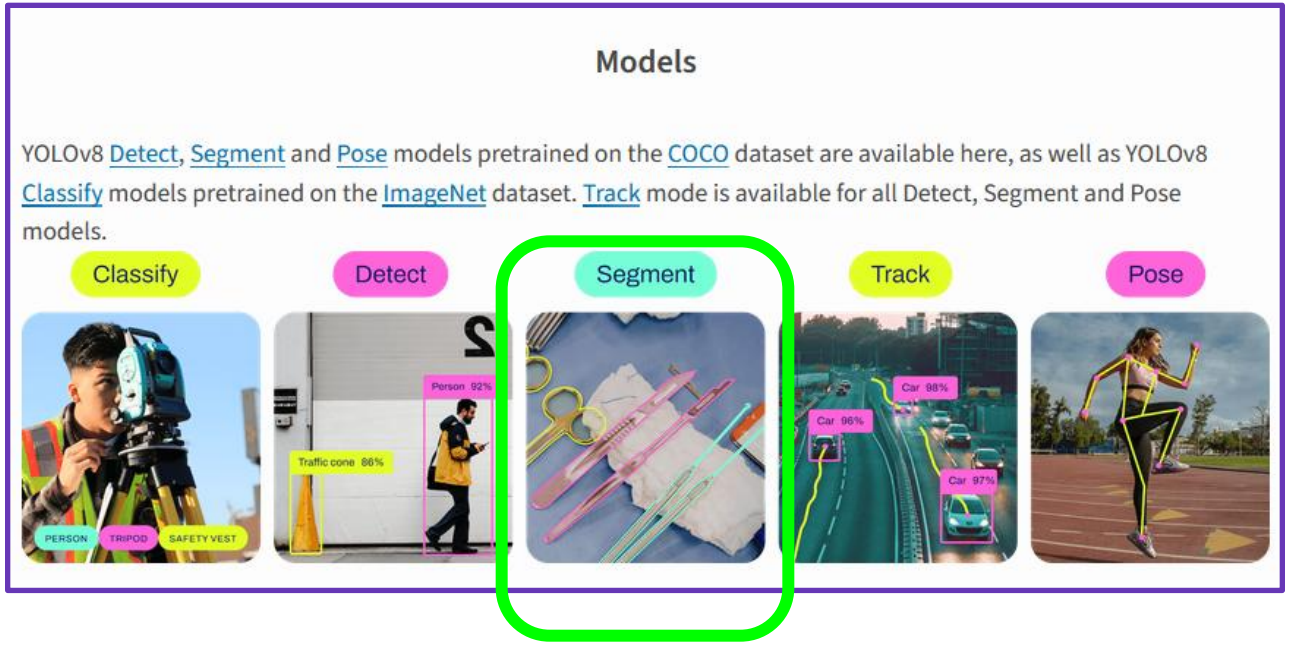


Fig.: Screenshot from https://pypi.org/project/ultralytics/ .

Sensor Fusion
Camera based object detection

Fig.: We want to use a precise object detection, so we go for a semantic segmentation.

Original screenshot from https://pypi.org/project/ultralytics/



Evaluation of an 3D object detector
Dataset and algorithm to be used

The original KITTI dataset as seen on the right does provide 3D bounding boxes (BBs).

We are going to use data of all sensors including the 3D BBs for this task!

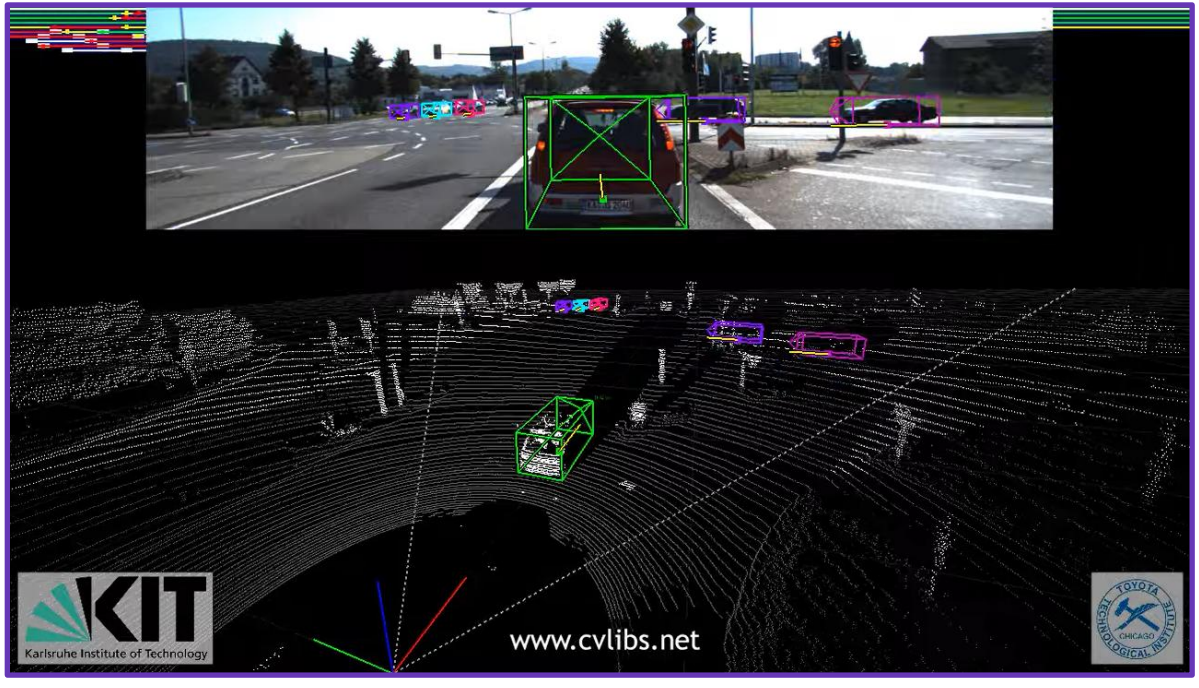


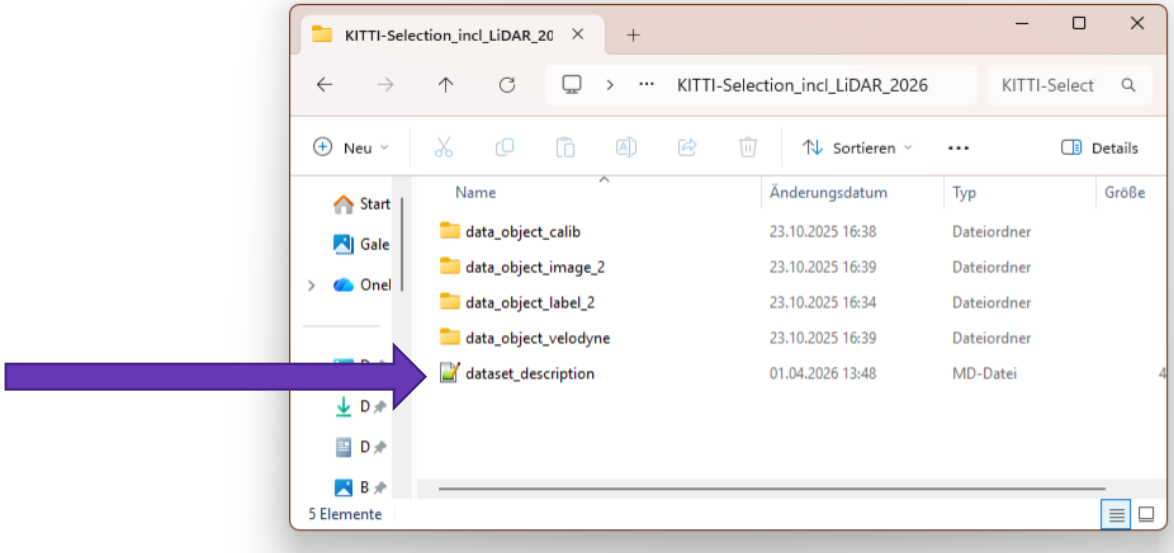
Fig.: Scene with labelled objects from the original KITTI dataset.
Source: https://www.cvlibs.net/datasets/kitti/index.php

Evaluation of an 3D object detector

Dataset and algorithm to be used

Fig.: The last file in the folder is the most important file.

Read the file dataset_description to get the needed information how to use the data provide as well as a link to the full official KITTI dataset.

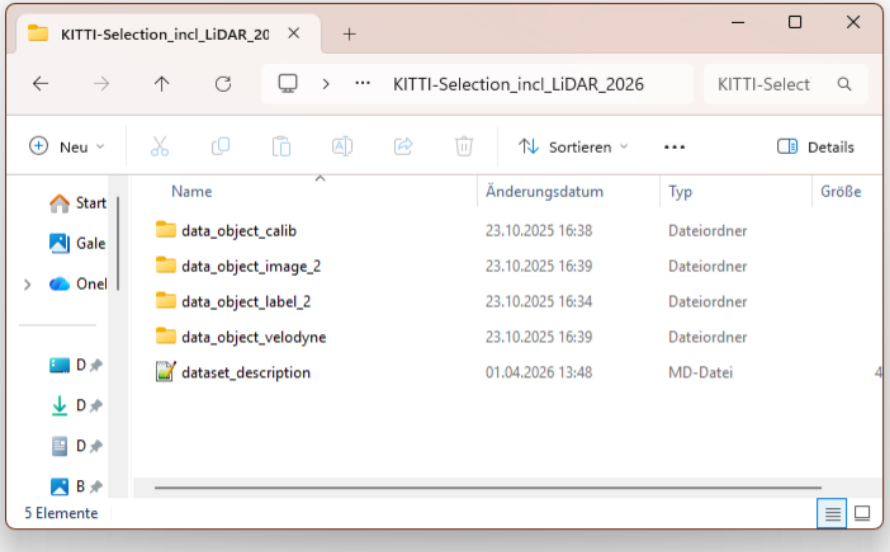


Evaluation of an 3D object detector

Dataset and algorithm to be used

Fig.: We did select some 20 frames from the original dataset put kept the notation as well as the folder structure as it is in the original dataset so that you can use helper function provided by the KITTI team.

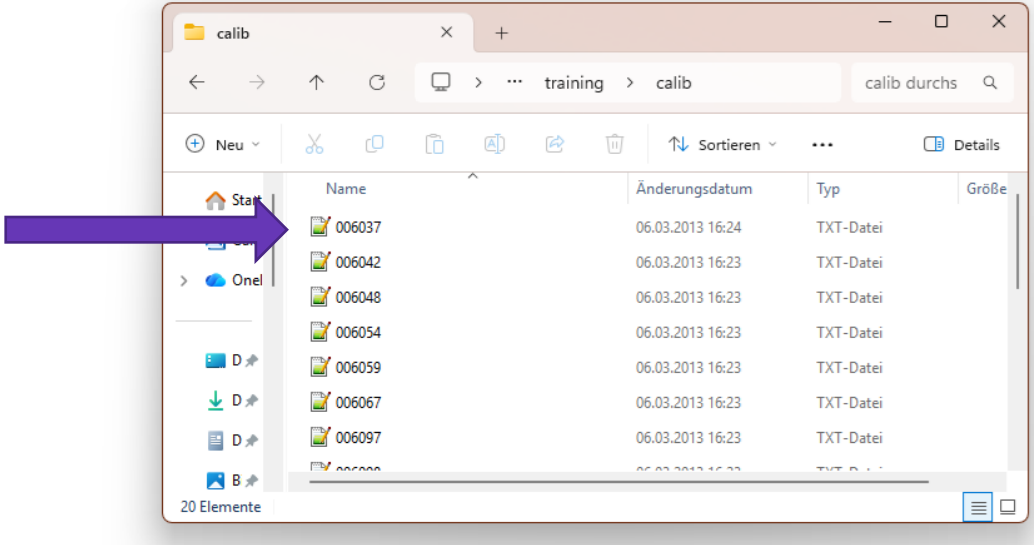
Click through the folders to understand quickly the structure.



Evaluation of an 3D object detector

Dataset and algorithm to be used

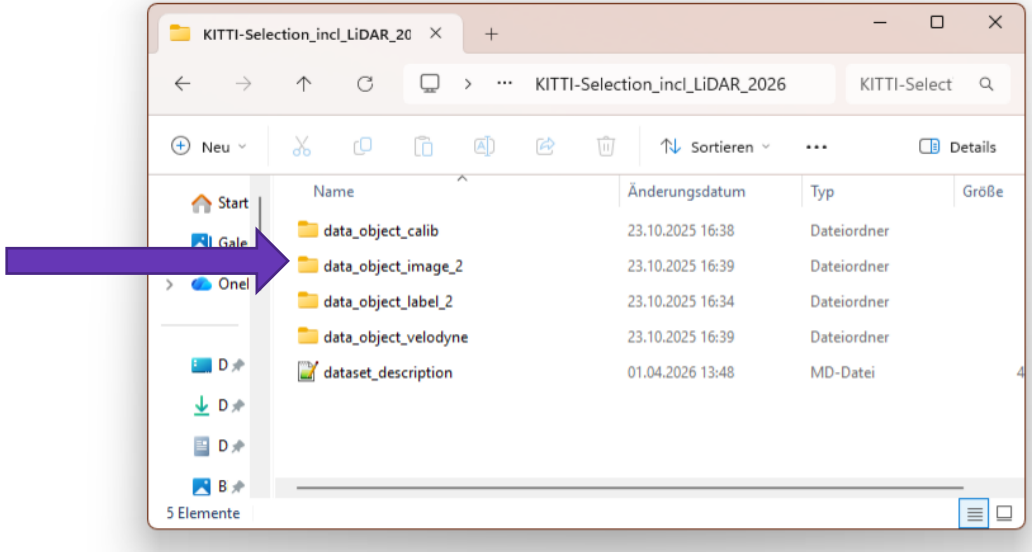
Fig.: For each of the 20 selected scenes there is the corresponding calibration information provides. The ID (e.g. 006037) will be the same for all data in this scene.



Evaluation of an 3D object detector
Dataset and algorithm to be used

Fig.: The next folder includes the camera images.

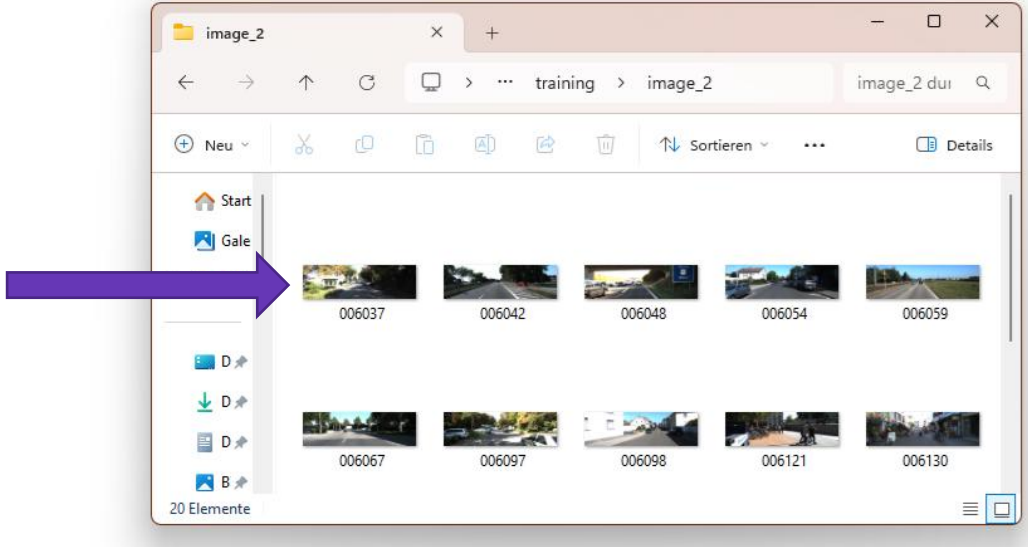
For this task, we will use the left top camera so the images provided by the other cameras is omitted in our subselection.



Evaluation of an 3D object detector
Dataset and algorithm to be used

Fig.: Again, you can find the scene with the corresponding ID, here again 006037.

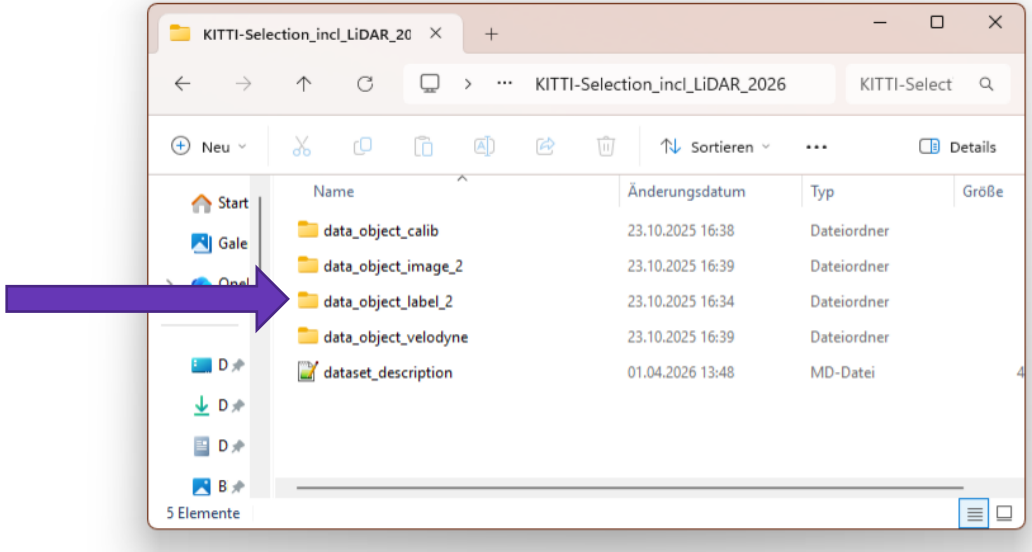
In the folder, you have access to the image of each scene.



Evaluation of an 3D object detector
Dataset and algorithm to be used

Fig.: The next folder includes the labeled data in the form of 3D bounding boxes (BBs).

For this task, we will use the provided 3D BBs to evaluate or detection algorithm based on lidar and camera data.

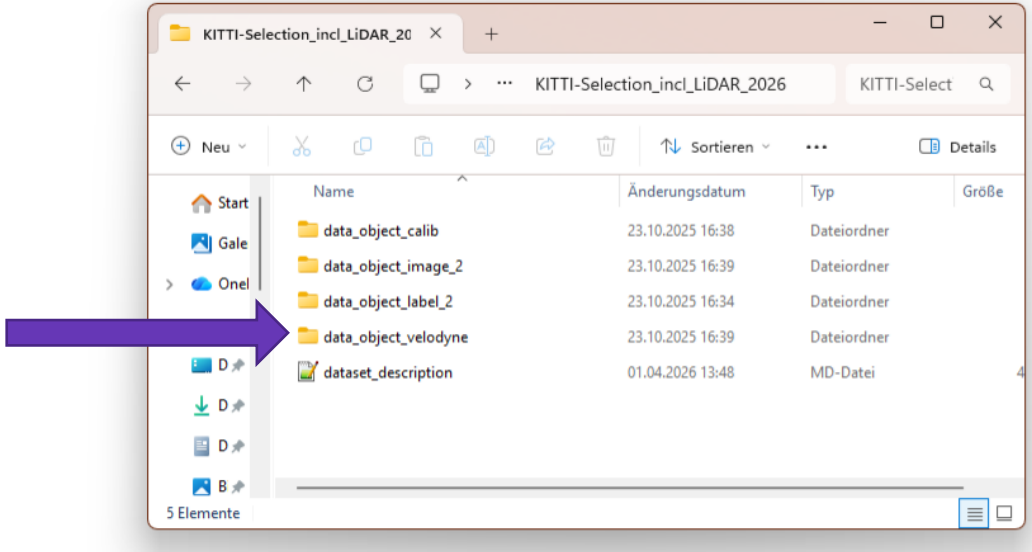


Evaluation of an 3D object detector

Dataset and algorithm to be used

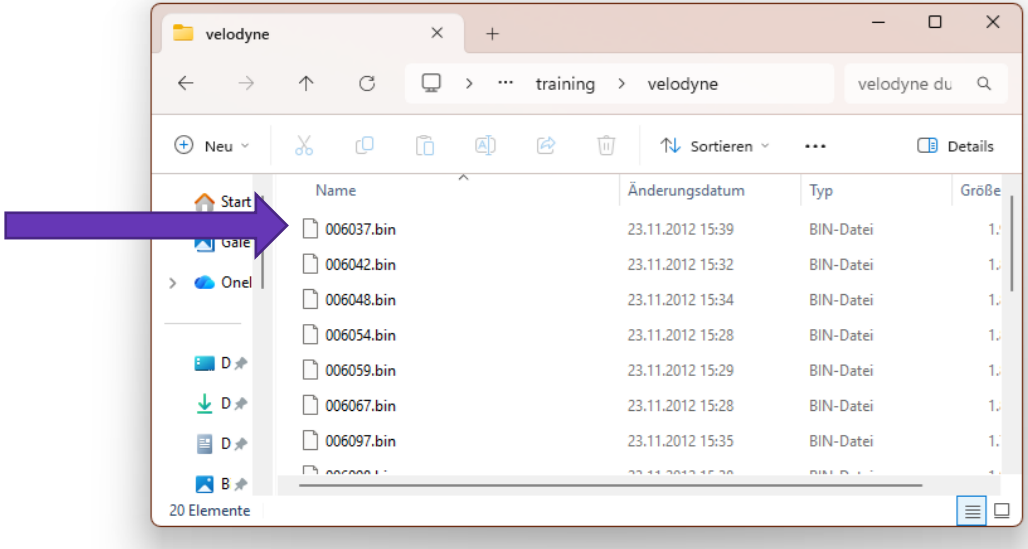
Fig.: The next folder includes the lidar data, here provided by a Velodyne lidar system.

For each scene you will find a full point cloud.



Evaluation of an 3D object detector
Dataset and algorithm to be used

Fig.: For each scene you will find a full pointcloud, here ID 00637.



Each point in this cloud consist of the 3D position and the intensity value. So you could consider this even a 4D point cloud.

As a first step, you should define a helper function to load and visualize this point cloud.

Evaluation of an 3D object detector

Dataset and algorithm to be used



Fig.: Now, you should be able to find this scene in the camera folder. Using the helper function, you should also be able to access the corresponding 3D point cloud.

Evaluation of an 3D object detector

Dataset and algorithm to be used

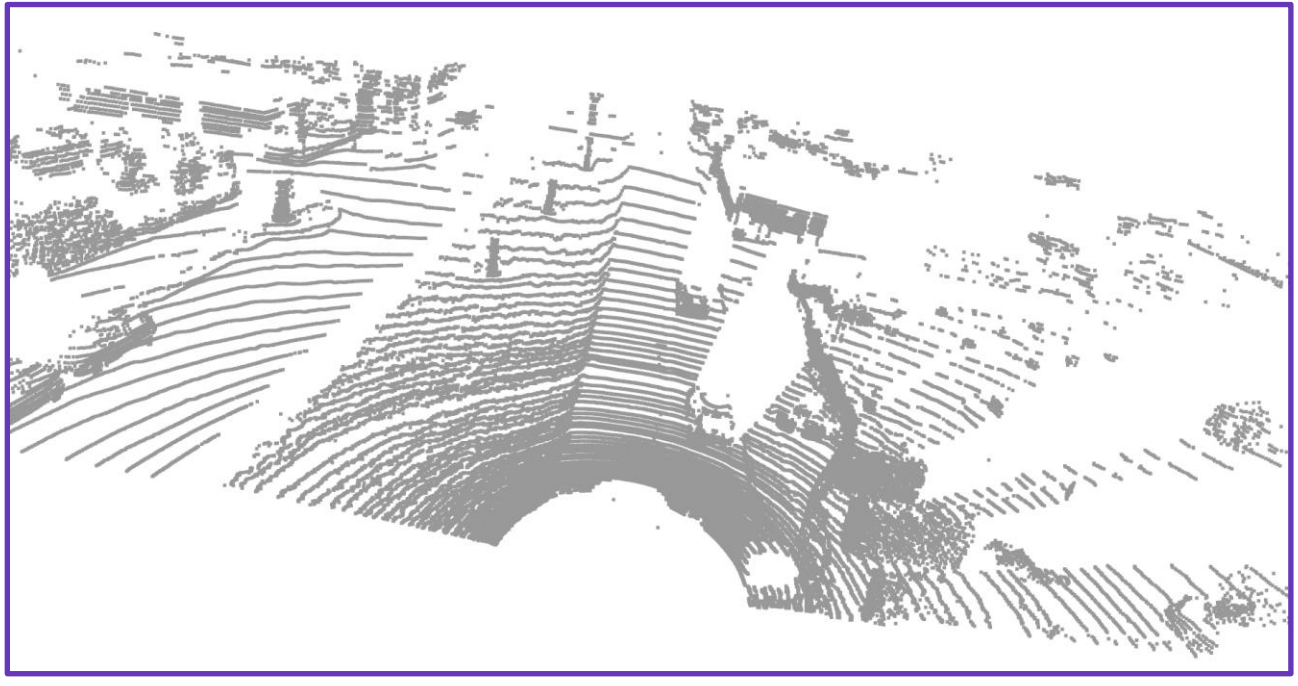


Fig.: The same scene now as 3D point cloud. But how can we know which point belongs to what car?

Evaluation of an 3D object detector

Dataset and algorithm to be used



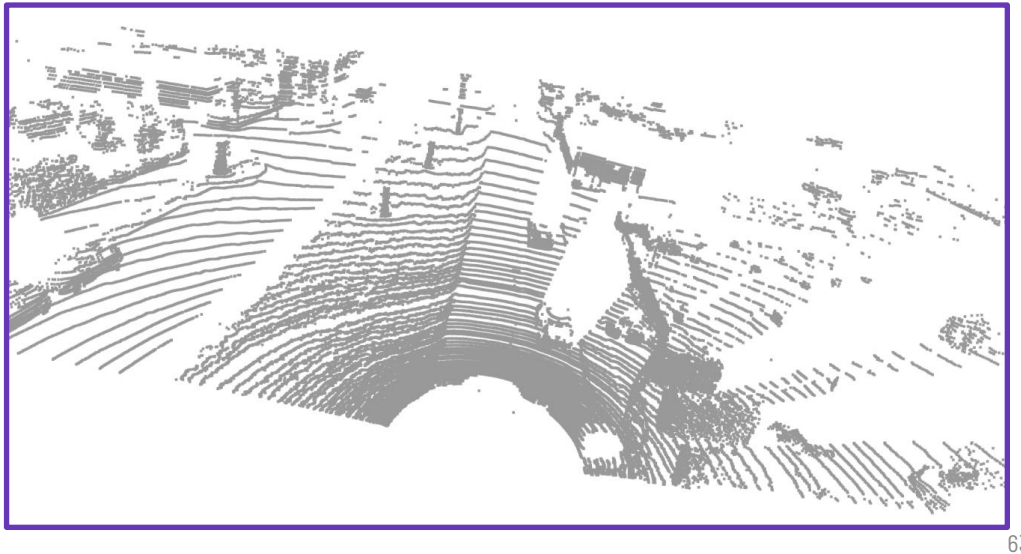


Fig.: Same scene recorded with different sensors.

We do see the cars in the image, but where are they in the point cloud? Can we somehow identify that?

Evaluation of an 3D object detector

Dataset and algorithm to be used



Fig.: Same scene recorded with different sensors.

We do see the cars in the image, but where are they in the point cloud? Can we somehow identify that?



Evaluation of an 3D object detector

What we have to do

We already know how to detect objects in camera images. For this, we again use the ultralytics YOLO but now we use the sematic instance segmentation to detect objects on a pixelwise level.

This can be used quite easily by pip install:

https://pypi.org/project/ultralytics/

https://docs.ultralytics.com/

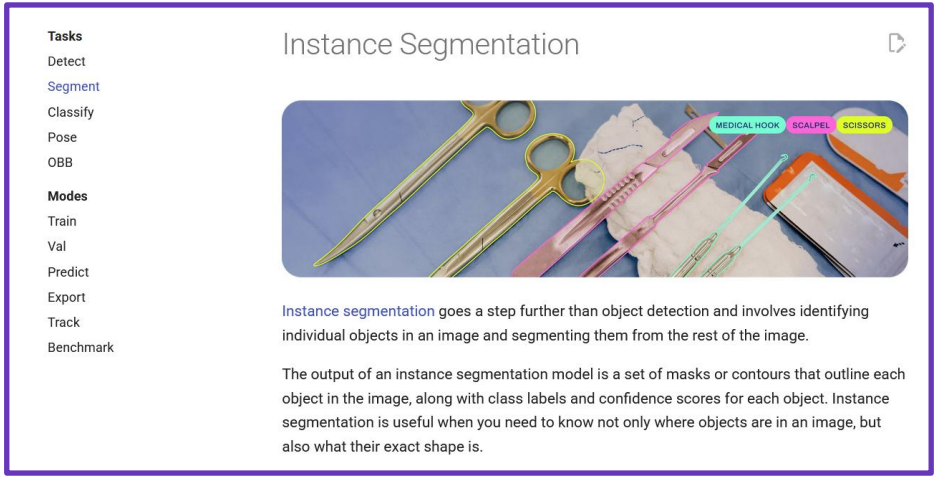


Fig.: Screenshot from https://docs.ultralytics.com/tasks/segment/

Evaluation of an 3D object detector

What we have to do

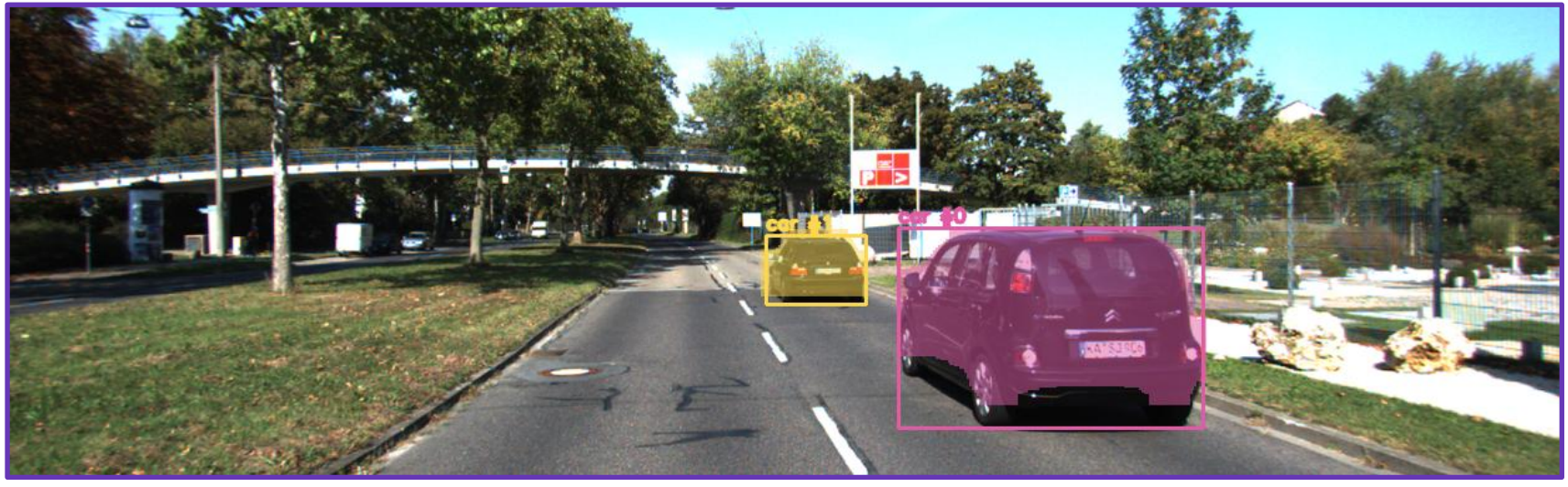


Fig.: Output of a YOLO semantic instance segmentation. Note that we now have not just the bounding boxes but also the pixelwise labeling available. Now, we can fuse the camera information with the lidar point cloud. **We do this for the cars only!**

Evaluation of an 3D object detector

What we have to do



Fig.: Lidar Point cloud projected into the image. Using the result of the semantic segmentation, we can decide which point belongs to what car object!

Evaluation of an 3D object detector

What we have to do

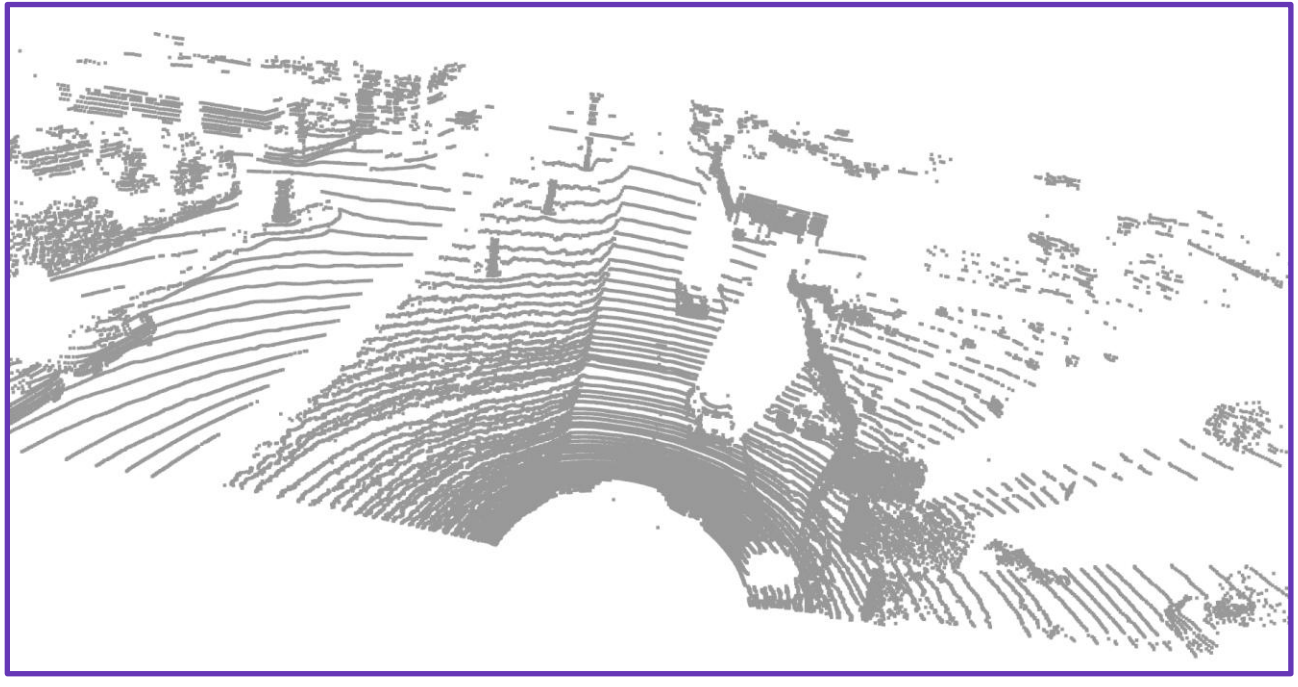


Fig.: The same scene as 3D pointcloud.

Evaluation of an 3D object detector

Dataset and algorithm to be used



Fig.: Using the projected points, we should be able to identify which points belong to what object....



Evaluation of an 3D object detector

What we have to do

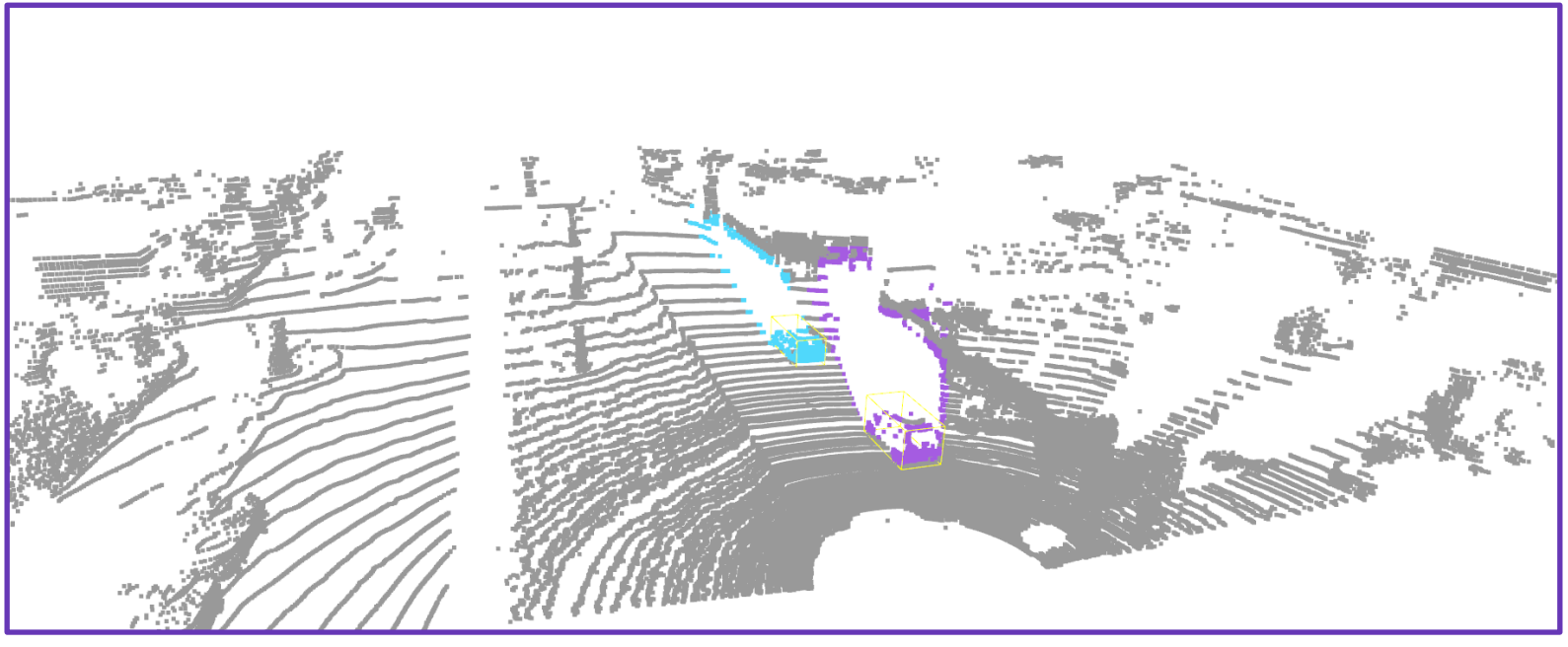


Fig.: The corresponding sub-pointclouds marked in color. So we know where the cars are placed in the 3D environment. **We now have a 3D object detector running!**

Evaluation of an 3D object detector

What we have to do

So we have a 3D object detector running on our machine.

But how good is it?



Evaluation of an 3D object detector

What we have to do

We can see that we have too many points per object available.

But which points are correctly taken as the sub-pointcloud for each detection and which are incorrectly included in the pointcloud?

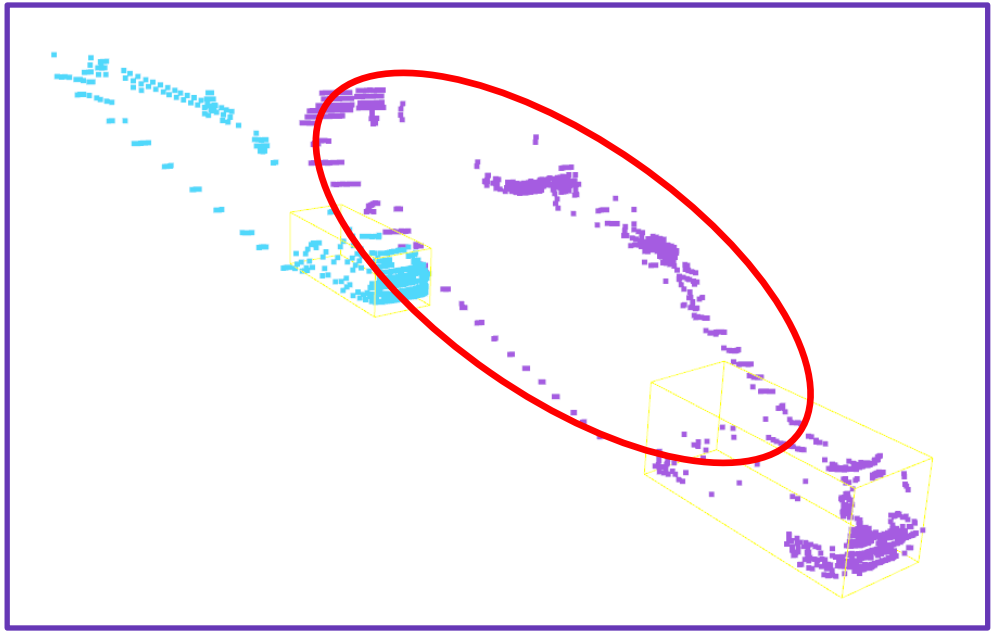


Evaluation of an 3D object detector

What we have to do

You can see this best on the object on the right. Here, we have a lot of incorrect points marked in red. This is called a “bleed-out”.

We now want to evaluate how good or bad our detector actually worked.



Evaluation of an 3D object detector

What we have to do

For this step, we will use the 3D bounding boxes provided by the labelled ground truth.

Only the points inside the corresponding bounding box can be considered to be classified correctly as belonging to the corresponding detected object.



All point outside are wrong.

Evaluation of an 3D object detector

What we have to do

For this task, we will not evaluate the performance of the YOLO sematic instance segmentation but focus on the evaluation of the overall 3D object detector.



Evaluation of an 3D object detector

What we have to do

To be able to evaluate and even improve your detector, you need to be able to answer the following questions:

- How many points are inside or outside the 3D BBs?
- Is the absolute amount of points important or the relative amount?
- Which detected object can then be considered a correctly detected object?
- How can we get rid of the “bleed-out” points? Does this improve our detection?



Evaluation of an 3D object detector
What we have to do

We also remember that **we originally wanted to estimate the distance of an object**. Using the sub-pointclouds, can we now truly calculate the distance of a car object!

For this you need to do the following:

- Calculate the distance to the car using your sub-pointcloud
- Calculate the actual distance using the ground-truth
- Compare the two distances



Evaluation of an 3D object detector
Goals

Evaluate the results of your object detector on the dataset provided via moodle.
Evaluate on the car class only.

Goals of this task:

- Calculate the sub-pointclouds of each car detection as described above
- Evaluate and improve your 3D object detector as described above
- Calculate the distance for each car and compare this to the ground-truth
- Make sure to fully understand the process you have used.

WHAT IS NEXT?

What is next?

What is next? Master Thesis!Moving on to a bigger Data Set

As part of a scientific project, you could move now fully to a larger data set like KITTI or the Astyx dataset. On this dataset, you could evaluate other object detectors as well.

The KITT leaderboard could be your starting point to select such a detector:

https://www.cvlibs.net/datasets/kitti/eval_object.php?obj_benchmark=3d

This would / could be your first steps into the direction on applied deep learning on sensor data!

What is next?
Finding the optimal fusion

If you are interested in further working on sensor fusion, the implementation and evaluation of your algorithm would be a very good starting point.

Of course there are different approaches to combine the data. In principle, these approaches can be seen as **three different types**:

- low level fusion
- feature level fusion
- high level fusion

What is next?
Finding the optimal fusion

At the moment, there is a lot of research going on with the goal to find an optimal sensor fusion.

If you are interested in this topic, you could do your own research as part of a scientific project or your master thesis.

A starting point could be one of the datasets discussed during the lecture. You will find a lot of material how to work with them online…

Your Own Stuff

What did you have in mind?

You always wanted to built a autonomous robot?

We have some robots available as well as several sensors (camera and lidar).

You could use these parts for your project as well!













Your Own Stuff
What did you have in mind?

Also possible would be your own software project.

We can discuss your personal ideas and see if they work as an individual project!

Just send me an email stefan.elser@rwu.de and we can meet to discuss your own project idea!

HOCHSCHULE
RAVENSBURG-WEINGARTEN
UNIVERSITY
OF APPLIED SCIENCES

VIELEN DANK FÜR IHRE AUFMERKSAMKEIT!

Doggenriedstraße
88250 Weingarten

Postfach 3022
88216 Weingarten

www.rwu.de
info@rwu.de